

# Aberration correction in long GRIN lens-based microendoscopes for extended field-of-view two-photon imaging in deep brain regions

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## eLife Assessment

This **valuable** study builds on previous work by the authors by presenting a potentially key method for correcting optical aberrations in GRIN lens-based microendoscopes used for imaging deep brain regions. By combining simulations and experiments, the authors provide **convincing** evidence showing that the obtained field of view is significantly increased with corrected, versus uncorrected microendoscopes. Because the approach described in this paper does not require any microscope or software modifications, it can be readily adopted by neuroscientists who wish to image neuronal activity deep in the brain.

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## Abstract

Two-photon (2P) fluorescence imaging through gradient index (GRIN) lens-based endoscopes is fundamental to investigate the functional properties of neural populations in deep brain circuits. However, GRIN lenses have intrinsic optical aberrations, which severely degrade their imaging performance. GRIN aberrations decrease the signal-to-noise ratio (SNR) and spatial resolution of fluorescence signals, especially in lateral portions of the field-of-view (FOV), leading to restricted FOV and smaller number of recorded neurons. This is especially relevant for GRIN lenses of several millimeters in length, which are needed to reach the deeper regions of the rodent brain. We have previously demonstrated a novel method to enlarge the FOV and improve the spatial resolution of two-photon microendoscopes based on GRIN lenses of length < 4.1 mm (Antonini et al. eLife 2020). However, previously developed microendoscopes were too short to reach the most ventral regions of the mouse brain. In this

study, we combined optical simulations with fabrication of aspherical polymer microlenses through three-dimensional (3D) microprinting to correct for optical aberrations in long (length > 6 mm) GRIN lens-based microendoscopes (diameter, 500  $\mu$ m). Long corrected microendoscopes had improved spatial resolution, enabling imaging in significantly enlarged FOVs. Moreover, using synthetic calcium data we showed that aberration correction enabled detection of cells with higher SNR of fluorescent signals and decreased cross-contamination between neurons. Finally, we applied long corrected microendoscopes to perform large-scale and high precision recordings of calcium signals in populations of neurons in the olfactory cortex, a brain region laying approximately 5 mm from the brain surface, of awake head-tethered mice. Long corrected microendoscopes are powerful new tools enabling population imaging with unprecedented large FOV and high spatial resolution in the most ventral regions of the mouse brain.

## Introduction

High resolution 2P fluorescence imaging of the awake brain is a fundamental tool to investigate the relationship between the structure and the function of brain circuits<sup>1</sup>. Compared to electrophysiological techniques, functional imaging in combination with genetically encoded indicators allows monitoring the activity of genetically targeted cell types, access to subcellular compartments, and tracking the dynamics of many biochemical signals in the brain<sup>2</sup>. However, a critical limitation of multiphoton microscopy lies in its limited (< 1 mm) penetration depth in scattering biological media<sup>3</sup>.

Deep ventral regions of the brain, such as the olfactory cortex, the hypothalamus, and the amygdala play fundamental roles in controlling the processing of sensory information<sup>4</sup>, circadian rhythms<sup>5</sup>, hormone release<sup>5</sup>, decision making<sup>6</sup>, and adaptation of instinctive and motivational behavior<sup>6</sup>.

Since these ventral areas lay deeper than 4 mm down the brain surface in rodents and larger mammals (e.g., rats, cats, and marmosets), imaging these regions typically requires the use of light guides, which relay the focal plane of the microscope objective down to the target brain area. Since inserting light guides in the brain damages the tissue above the target area<sup>7</sup>, reducing the cross section of the probe is desired when imaging these deep brain regions. One popular solution for deep 2P imaging in awake animals is using GRIN lenses<sup>8–11</sup>, thin cylindrical rod lenses characterized by refractive index which varies with the radial distance<sup>12</sup>. For example, in the mouse orbitofrontal cortex 2P microscopy through GRIN endoscopes showed that orbitofrontal neurons respond to either caloric rewards or social stimuli, indicating the presence of functionally distinct feeding and social neuronal subnetworks in this deep cortical region<sup>13</sup>. Similarly, in the lateral hypothalamus combining GRIN endoscopes with 2P imaging revealed neurons encoding thermal punishment and reward and this subset of neurons was distinct from the ensemble of neurons encoding caloric reward<sup>14</sup>. Other ventral brain regions, such as the olfactory cortex, can be imaged with an invasive preparation, which does not require a GRIN lens but which is incompatible with imaging in awake behaving animals<sup>15, 16</sup>. In the olfactory cortex, GRIN lens-based 2P endoscopy becomes necessary in awake animals<sup>17</sup>, where head tilting is not compatible with animal performance in a behavioral task.

One limitation of GRIN lens-based endoscopy is that GRIN lenses suffer from intrinsic optical aberrations<sup>18, 19</sup>. GRIN aberrations can be both on-axis and off-axis and severely degrade the quality of 2P imaging. GRIN aberrations distort the point spread function (PSF) and enlarge the excitation volume of the focal spot, leading to reduced SNR of fluorescence signals and decreased spatial resolution<sup>7, 20, 21</sup>. This effect is not uniform across the FOV and is particularly relevant in lateral portions of the FOV, generating uneven spatial resolution and inhomogeneous amplitude of fluorescence signals across the FOV<sup>7, 20, 21</sup>. As a consequence of these

limitations, previous studies using 2P imaging through GRIN lenses in ventral regions of the mouse brain were limited to imaging a small number of neurons with uneven spatial resolution across the FOV and low SNR in lateral portion of the FOV <sup>13, 14, 22</sup>. Developing GRIN-based endoscopic probes with improved optical properties and more homogeneous spatial resolution, which are long enough to enable imaging in the most ventral regions of the rodent brain while maintaining a small ( $\leq 500\ \mu\text{m}$ ) cross section, is thus a compelling need.

Various adaptive optics methods were developed to correct optical aberrations in GRIN lenses. For example, initial efforts utilized optical compensation by low-order electrostatic membrane mirror <sup>18</sup>. Alternatively, adaptive optics through pupil segmentation was used to efficiently correct GRIN aberrations, increasing the intensity of recorded fluorescence signals and enlarging the imaging FOV <sup>20, 21</sup>. More recently, geometric transformation adaptive optics has been developed and applied for the correction of aberration in GRIN lenses <sup>23</sup>. However, adaptive optics approaches require substantial modification of the optical path of the microscope <sup>18, 20, 21, 23</sup>, needs specifically designed software controls <sup>20, 21, 23</sup>, and may limit the temporal resolution of imaging <sup>20, 21</sup>. Alternatively, mm-size plano-convex lenses have been combined with GRIN lenses to correct on-axis aberrations and increase the Numerical Aperture (NA) of the optical system <sup>11</sup>. However, this method is difficult to apply to GRIN lenses of cross section smaller than 1 mm <sup>24</sup>, because of the difficulty in manufacturing high-precision optics with small radial dimension and complex profile. Recently, a novel method based on 3D microprinting of polymer optics was developed to correct for GRIN aberrations by placing specifically designed aspherical corrective lenses at the back end of the GRIN lens <sup>7</sup>. This approach is attractive because it is built-in on the GRIN lens and corrected microendoscopes are ready-to-use, requiring no change in the optical set-up. However, previous work demonstrated the feasibility of this method only for GRIN lenses of length  $< 4.1\ \text{mm}$  <sup>7</sup>, which are too short to reach the most ventral regions of the mouse brain. The applicability of this technology to longer GRIN lenses, which are affected by stronger optical aberrations <sup>19</sup>, remained to be proven.

In this study, we designed, developed and validated correction of optical aberrations in GRIN lenses of length 6.4 mm and 8.8 mm and diameter 500  $\mu\text{m}$  using a combination of ray-trace simulation, 3D microprinting by 2P lithography, simulated calcium data, and experimental characterization. Validation of the new corrected microendoscopes is provided with proof-of-principle experiments in a ventral brain region, the olfactory cortex, of awake head-fixed mice.

## Results

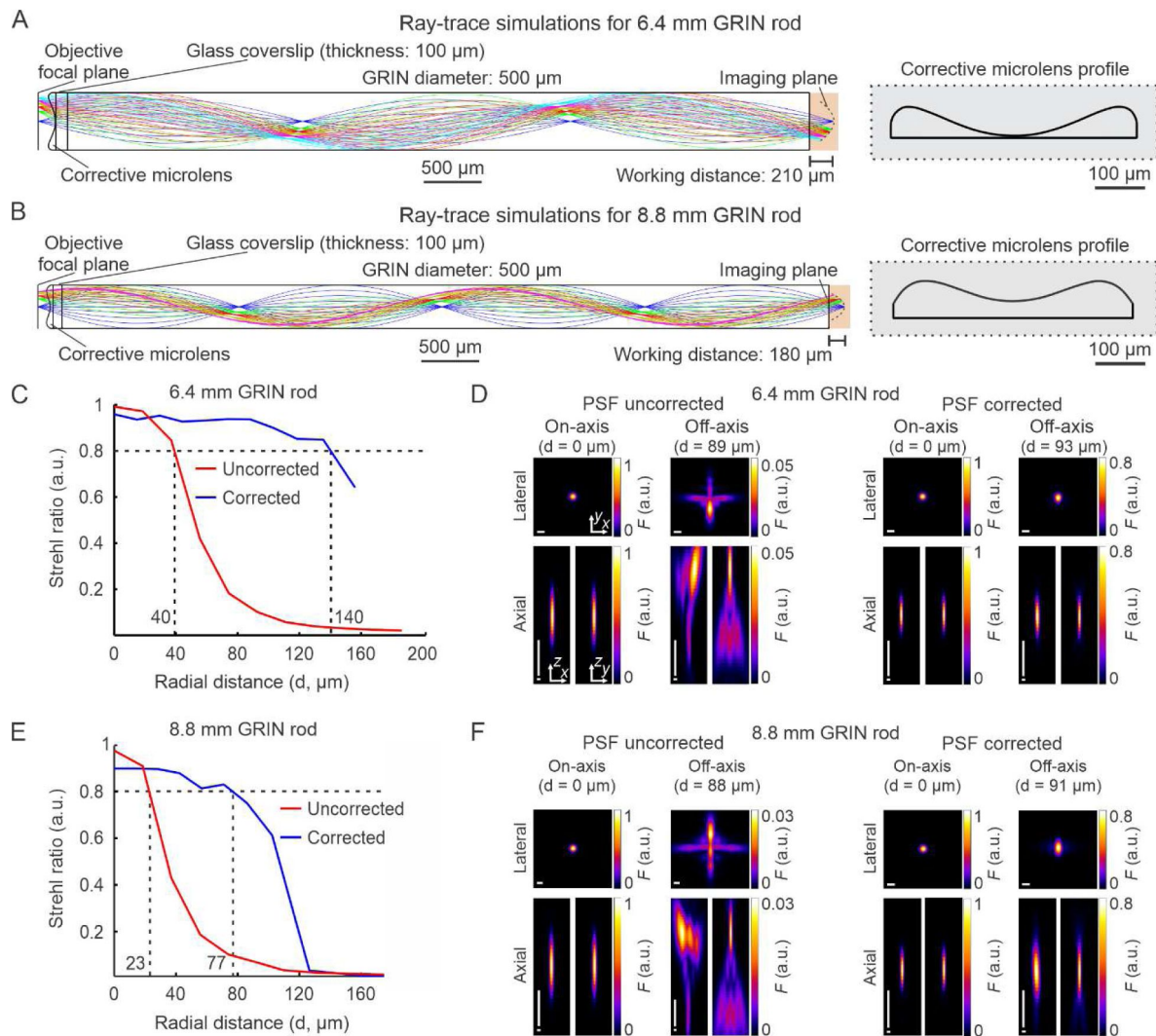
### Design of aberration corrected microendoscopic probes based on long GRIN rods

We selected two GRIN singlet rod lenses (here on GRIN rods) suitable for multiphoton fluorescence imaging in ventral regions of the mouse brain. The two selected GRIN rods had NA = 0.5 on both sides, 500  $\mu\text{m}$  diameter, 1.5 and 2.0 pitch, and 6.4 mm and 8.8 mm length, respectively. For both GRIN rods, we modeled an optical assembly composed of the GRIN rod, a thin layer of glass (representing a 100  $\mu\text{m}$  thick glass coverslip) attached to the back end of the GRIN rod, and an aspheric corrective lens (here on called corrective lens) aligned to the rod and attached to the opposite side of the glass coverslip with respect to the GRIN rod (**Fig. 1A, B**). We designed the aspheric surface of the corrective lens using ray-trace simulations at the wavelength ( $\lambda_{\text{exc}}$ ) of 920 nm, the excitation wavelength commonly used for 2P imaging of the popular genetically encoded calcium indicator GCaMP. The design of the corrective lens was an iterative process in which the parameters that define the corrective lens surface were automatically varied to obtain the highest and most homogeneous spatial resolution possible over the longest possible radial distance from the optical axis (see **Supplementary Table 1** and Materials and Methods for details on simulation parameters). More specifically, we simulated the corrective lens profile such that the

Strehl ratio (defined as the peak intensity of the focal spot of an aberrated optical system normalized to the peak intensity of the focal spot of the ideal optical system<sup>25</sup>) of the corrected microendoscope remained above the threshold of 0.8 for the longest possible radial distance from the optical axis. The values of Strehl ratio equal to 0.8 corresponded to the lower bound of the diffraction-limited condition set by the Maréchal criterion<sup>7,25</sup>. Our simulations showed that the profiles of the two corrective lenses displayed in **Fig. 1 A, B** led to an increase in the radius of the diffraction-limited FOV compared to the case in which the corrective lens was not positioned in the back end of the GRIN rods (uncorrected microendoscopes). The profile of the corrective optical element was specific to the type of GRIN rod considered. The simulated increase in the radius of the diffraction-limited FOV was 3.50 times and 3.35 times for the 6.4 mm-long and 8.8 mm-long probe, respectively (**Fig. 1C, E**). Moreover, we investigated the effect of changing wavelength on the Strehl ratio. We found that the Strehl ratio remained  $> 0.8$  within at least  $\pm 10$  nm from  $\lambda = 920$  nm (**Supplementary Figure 1**), which covered the limited bandwidth of our femtosecond laser. All simulations were performed maximizing aberration correction in the focal plane of the microendoscopes. However, we explored the effect of aberration correction outside the focal plane. In corrected endoscopes, we found that for off-axis rays (radial distance from the optical axis  $> 40$   $\mu\text{m}$ ) the Strehl ratio was  $> 0.8$  (Maréchal criterion) in a larger volume compared to uncorrected endoscopes (new **Supplementary Figure 2**), demonstrating that the aberration correction method developed in this study extends for short distances beyond the focal plane. For example, at a radial distance of  $\sim 90$   $\mu\text{m}$  from the optical axis, the axial range in which the Strehl ratio was  $> 0.8$  in corrected endoscopes was 28  $\mu\text{m}$  and 19  $\mu\text{m}$  for the 6.4 mm- and the 8.8 mm-long microendoscope, respectively. Decentering the optical axis of the corrective lens from the optical axis of the GRIN rod rapidly disrupted the optical properties of the corrected microendoscopes (**Supplementary Figure 3**). To visualize the expected improvement of the spatial resolution in corrected microendoscopes, we simulated the PSF of uncorrected and corrected microendoscopes at different radial distances from the optical axis. For both the 6.4 mm-long and 8.8 mm-long GRIN rods, we found that the PSF of the uncorrected probe was strongly aberrated and highly irregular at  $\sim 90$   $\mu\text{m}$  away from the center of the FOV (**Fig. 1D, F**). In contrast, in the corrected microendoscopes the PSF had regular shape and, at  $> 90$   $\mu\text{m}$  away from the optical axis, it was similar to the one measured at the center of the FOV (**Fig. 1D, F**, see also **Supplementary Table 2**).

## Microfabrication of corrective lenses and assembly of long corrected microendoscopes

Due to the small size of the simulated corrective lenses (diameter: 500  $\mu\text{m}$ ; height: few tens of  $\mu\text{m}$ ), we first used 3D microprinting based on 2P lithography to fabricate aspherical corrective lens prototypes (**Fig. 2A, B**)<sup>26,27</sup>. We then replicated the prototypes using a molding procedure<sup>28</sup> (**Fig. 2C**). The molding strategy enabled us to obtain large numbers of aspherical lens replica directly manufactured onto a round glass coverslip in a much faster and cheaper way compared to 3D microprinting. We checked that the surface of the corrective lenses was in good agreement with the simulated profile (**Supplementary Figure 4**). Next, we assembled microendoscopes following the scheme in **Fig. 2D**, using a plain coverslip for the uncorrected case, while we used a coverslip with a corrective lens replica for the corrected case (this applies to all the experiments described in the manuscript). In the assembly of the microendoscope, we attached the round coverslip to the annular support structure shown in (**Fig. 2D**, called hereon ring), which had two functions: *i*) to facilitate holding the probe during stereotactic surgery implantation; *ii*) to provide stability to the microendoscope once implanted on the animal skull. We used either metallic rings (external diameter: 7 mm) or plastic rings (external diameter: 4.5 mm). Plastic rings were specifically designed and 3D printed to be compatible with commercial holders (ProView Implant Kit, Inscopix Inc. Mountain View, CA) to ease stereotaxic implantation (see Materials and Methods).

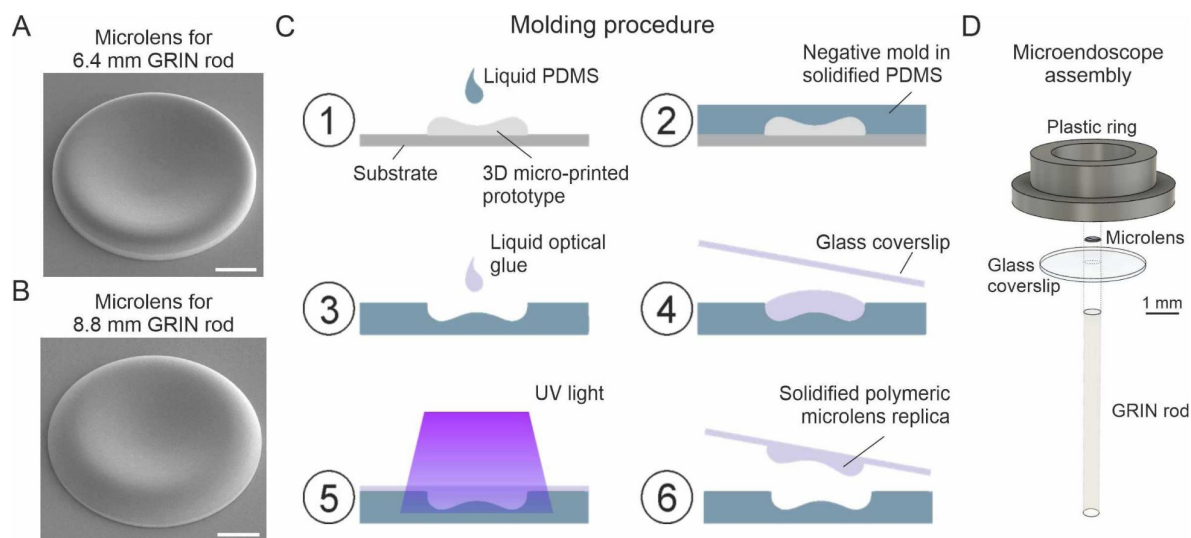


**Figure 1.**

### Optical simulations of long corrected microendoscopes.

A) Ray-trace simulation for the microendoscope based on the 6.4 mm-long GRIN rod. Left: rays of 920 nm light are relayed from the objective focal plane to the imaging plane. Labels indicate geometrical parameters of the microendoscope components. Right: profile of the corrective aspherical lens that maximizes the microendoscope FOV in the optical simulation shown on the left. B) Same as (A) for the microendoscope based on the 8.8 mm-long GRIN rod. C,D) Optical performance of simulated microendoscope based on the 6.4 mm-long GRIN rod. C) Strehl ratio as a function of the field radial distance (zero indicates the optical axis) computed on the focal plane in the object space (after the GRIN rod) for the corrected (blue) and the uncorrected (red) microendoscope. The black horizontal dashed line indicates the diffraction-limited threshold according to the Maréchal criterion <sup>25</sup>. The black vertical dashed lines mark the abscissa values of the intersections between the curves and the diffraction-limited threshold. D) Lateral ( $x,y$ ) and axial ( $x,z$  and  $y,z$ ) intensity profiles of simulated PSFs on-axis (distance from the center of the FOV  $d = 0 \mu\text{m}$ ) and off-axis (at the indicated distance  $d$ ) for the uncorrected (left) and the corrected microendoscope (right). Horizontal scale bars: 1  $\mu\text{m}$ ; vertical scale bars: 10  $\mu\text{m}$ . E,F) Same as (C,D) for the microendoscope based on the 8.8 mm-long GRIN rod.





**Figure 2.**

### **Fabrication of corrective lenses using 3D microprinting and assembly of long microendoscopes.**

A) Scanning electron microscopy image of the 3D microprinted replica of the corrective lens for the 6.4 mm-long corrected microendoscope. Scale bar: 100  $\mu\text{m}$ . B) Same as (A) for the 8.8 mm-long microendoscope. C) Molding procedure for the generation of corrective lens replica. Freshly prepared PDMS is casted onto a corrective aspherical lens printed with 2P lithography (1). After 48 hours, the solidified PDMS provides a negative mold for the generation of lens replica (2). A small drop of optical UV-curable glue is deposited onto the mold (3). The mold filled by glue is covered with a glass coverslip, which is gently pressed against the mold (4). The optical glue is polymerized with UV light (5). The coverslip with the attached polymeric lens replica is detached from the negative mold (6). Object dimensions are not to scale. D) Exploded view of the corrected microendoscope assembly.

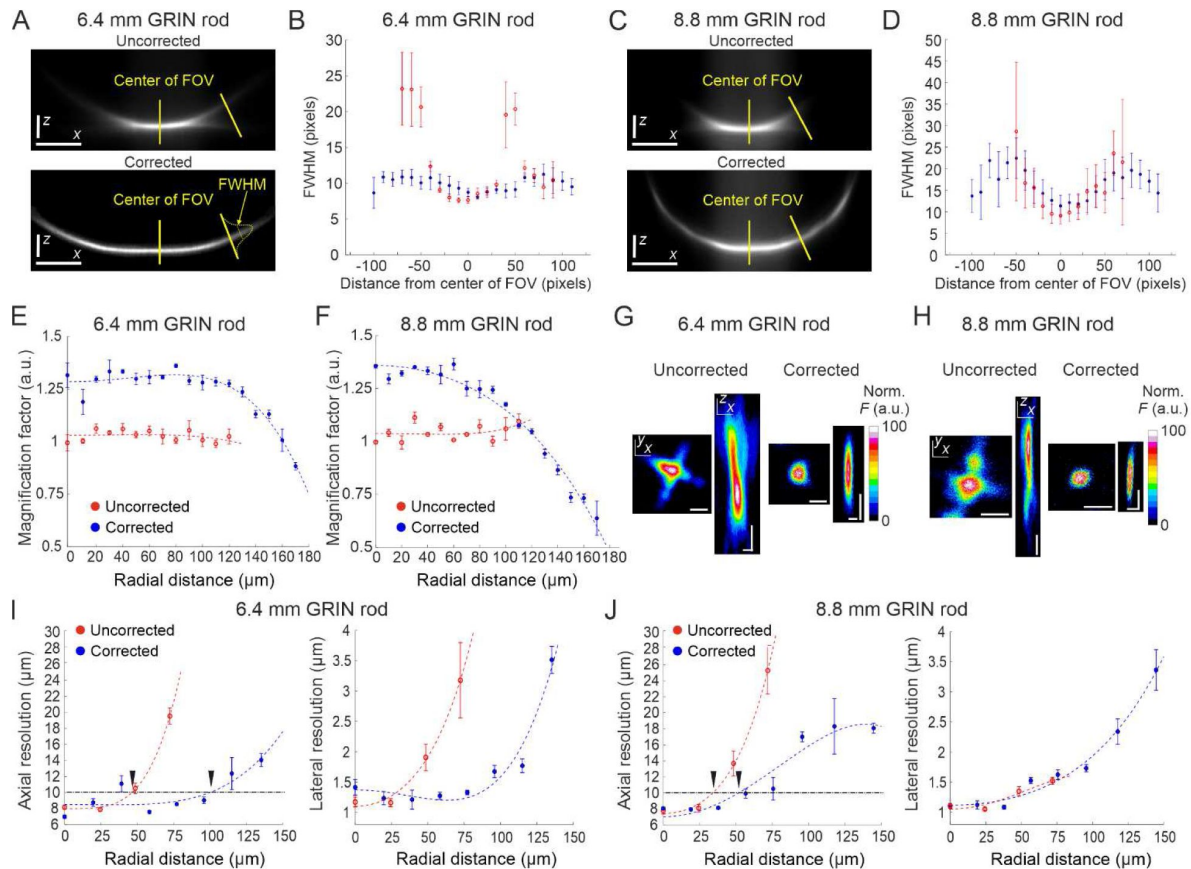
The alignment and assembly of all the microendoscope components was performed with a custom optomechanical set-up similar to that described in [7](#).

## Improved optical performance of long corrected microendoscopes

We tested the optical performance of the new corrected microendoscopes in 2P laser scanning microscopy (2PLSM) using a pulsed laser tuned at 920 nm. Microendoscopes were coupled to the 2P microscope using a previously described customized mount [7](#). To visualize the effect of the corrective lens on the axial extension of the excitation volume, we imaged subresolved homogenous fluorescent layers (layer thickness: ~300 nm) using uncorrected and corrected microendoscopes. **Figure 3A-D** shows the *x,z* intensity projections of *z*-stacks acquired through the subresolved fluorescent layers under both conditions (i.e., uncorrected and corrected microendoscopes). Strong off-axis aberrations of the GRIN lens generated a significant loss of fluorescence intensity in the marginal portions of the FOV in uncorrected microendoscopes (both the 6.4 mm-long, **Fig. 3A, B**, and the 8.8 mm-long, **Fig 3C, D**). In contrast, fluorescence signal was collected over longer distances from the center of the FOV in corrected microendoscopes (**Fig. 3A-D**). Moreover, for GRIN rods of both lengths the full width at half maximum (FWHM) of the film profile obtained with uncorrected microendoscopes increased with distance from the optical axis and became rapidly larger than 20  $\mu\text{m}$ . In contrast, the film profile obtained with corrected microendoscopes showed more homogenous FWHM values across an extended FOV (**Fig. 3A-D**).

To evaluate the increase in area of the sample that could be imaged with the corrected microendoscopes, we first characterized potential distortion of the FOV introduced by the microendoscopes. Using a calibration ruler, we measured the local magnification factor at different radial distances from the optical axis (**Fig. 3E, F**). We defined the local magnification factor as the ratio between the nominal pixel size (i.e. pixel size that applies to undistorted images collected with the microscope objective alone) and the real pixel size of the image collected with the same microendoscope objective coupled to the microendoscope (see Materials and Methods). Pixel size and distance displayed in the following panels and graphs are calibrated for FOV distortion and indicate real dimensions in the samples (see also Materials and Methods and **Supplementary Table 3**). We then measured the spatial resolution of microendoscopes by imaging subresolved fluorescent beads (bead diameter: 100 nm) located at different radial distances from the optical axis (**Fig. 3G, J**). We found that the axial dimension of the beads imaged with uncorrected microendoscopes rapidly increased with radial distance, whereas the spatial resolution of corrected microendoscopes remained more homogeneous over longer radial distances and more similar to the one measured on-axis (**Fig. 3G, J** and **Supplementary Table 4**). We set a threshold of 10  $\mu\text{m}$  on the axial resolution to define the radius of the effective FOV (corresponding to the black triangles in **Fig. 3I, J**) in uncorrected and corrected microendoscopes. We observed a relative increase of the effective FOV radius of 2.17 and 1.53 for the 6.4 mm- and the 8.8 mm-long microendoscope, respectively (**Table 1**). This corresponded to an enlargement of the effective FOV area of 4.7 times and 2.3 times for the 6.4 mm-long microendoscope and the 8.8 mm-long microendoscope, respectively (**Table 1**). These findings were in agreement with the results of the ray-trace simulations (**Figure 1**) and the measurement of the subresolved fluorescence layers (**Figure 3A-D**).

The optical characterization described above indicated that corrected microendoscopes of both lengths enabled imaging a larger FOV area compared to uncorrected microendoscopes. This result can be explained by the fact that corrected microendoscopes showed higher spatial resolution and increased probability of 2P fluorescence excitation in the marginal portions of the FOV. Corrected microendoscopes should therefore allow highly contrasted imaging of cell bodies and thin cellular processes over an extended FOV compared to uncorrected microendoscopes. Using both uncorrected and corrected microendoscopes, we confirmed these predictions by imaging the same region of a fixed brain slice expressing jRCaMP7f in neurons. We found that aberration correction



**Figure 3.**

### Optical characterization of long corrected microendoscopes shows improved spatial resolution over an enlarged FOV.

A) Representative x (horizontal), z (vertical) projection of a z-stack of a subresolved fluorescent layer acquired with the uncorrected (top) or corrected (bottom) microendoscope based on the 6.4 mm-long GRIN rod.  $\lambda_{\text{exc}} = 920$  nm; scale bars: 50 pixels. B) Thickness (mean values  $\pm$  standard error of the mean (s.e.m.)) of the layer as a function of the distance from the center of the FOV for uncorrected (red,  $n = 4$ ) or corrected (blue,  $n = 4$ ) microendoscopes. The thickness of the film is measured as the FWHM of the Gaussian fit of the fluorescence intensity along segments orthogonal to the tangential line to the section of the film and located at different distances from the center of the FOV (see yellow labels on the bottom image). C,D) Same as (A,B) for the microendoscope based on the 8.8 mm-long GRIN rod. E) The distortion of the FOV in uncorrected and corrected microendoscopes is evaluated using a calibration ruler. The magnification factor is defined as the ratio between the nominal and the real pixel size of the image and shown as a function of the radial distance for uncorrected (red,  $n = 3$ ) or corrected (blue,  $n = 3$ ) microendoscopes. Data are shown as mean values  $\pm$  s.e.m.. Fitting curves are quartic functions  $f(x) = ax^4 + bx^2 + c$  (see also Supplementary Table 3 for details). F) Same as (E) for the microendoscope based on the 8.8 mm-long GRIN rod. G-J) The spatial resolution of microendoscopes was measured acquiring z-stacks of subresolved fluorescent beads (bead diameter: 100 nm) located at different radial distances using 2PLSM ( $\lambda_{\text{exc}} = 920$  nm). G) Representative x,y and x,z projections of a fluorescent bead located at a radial distance of 75  $\mu\text{m}$ , imaged through an uncorrected (left) or a corrected (right) 6.4 mm-long microendoscope. Horizontal scale bars, 2  $\mu\text{m}$ ; vertical scale bars, 5  $\mu\text{m}$ . H) Same as G) for the microendoscope based on the 8.8 mm-long GRIN rod. I) Axial (left) and lateral (right) resolution (i.e. average size of the x,z and x,y projections of imaged beads, respectively) as a function of the radial distance from the center of the FOV for uncorrected (red) and corrected (blue) probes. Each data point represents the mean value  $\pm$  s.e.m. of  $n = 4$ -24 beads imaged using at least  $m = 3$  different 6.4 mm-long microendoscopes. Fitting curves are quartic functions  $f(x) = ax^4 + bx^2 + c$  (see Supplementary Table 4 for details). The horizontal black dash-dotted line indicates the axial resolution threshold of 10  $\mu\text{m}$ . The black triangles indicate the intersections between the threshold and the curves fitting the data and mark the estimated radius of the effective FOV of the probes. J) Same as (I) for the microendoscopes based on the 8.8 mm-long GRIN rod.



| Microendoscope type  | Effective FOV radius ( $\mu\text{m}$ ) | Fold increase in FOV radius | Fold increase in FOV area |
|----------------------|----------------------------------------|-----------------------------|---------------------------|
| 6.4 mm-long GRIN rod | Uncorrected: 46<br>Corrected: 100      | 2.17                        | 4.7                       |
| 8.8 mm-long GRIN rod | Uncorrected: 34<br>Corrected: 52       | 1.53                        | 2.3                       |

**Table 1.**

**Experimental measurement of enlarged effective FOV in long corrected microendoscopes.**

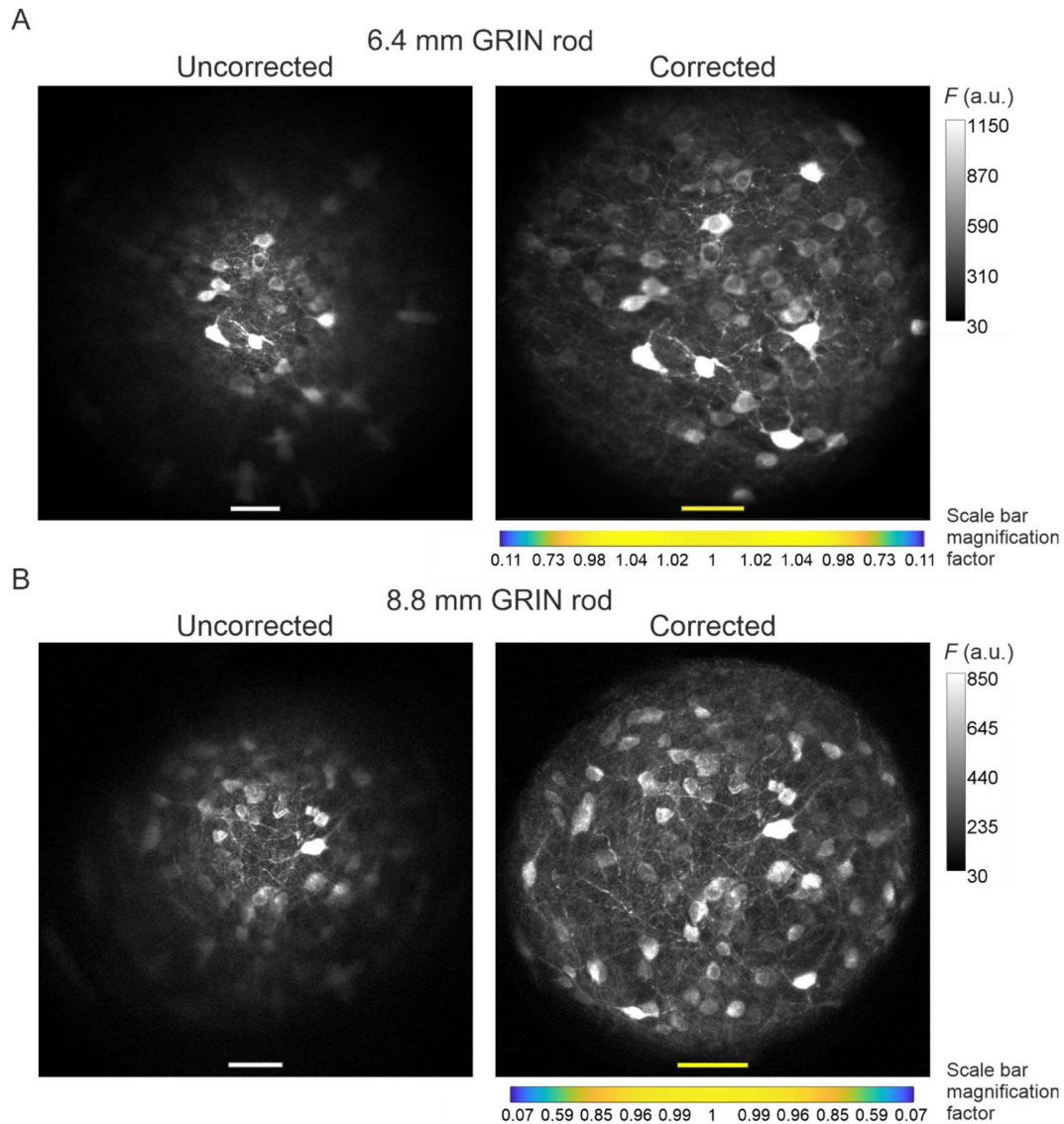
The values of the effective FOV radius for uncorrected and corrected microendoscopes were estimated from the intersection between the arbitrary threshold of 10  $\mu\text{m}$  on the axial resolution and the quartic function fitting the experimental data of **Fig. 3I, J** [↗](#).

enabled to clearly resolve neuronal somata and processes in the lateral portions of the FOV, whereas in the absence of the corrective lens the same structures located in the peripheral portion of the FOV appeared dim, strongly aberrated, and hardly detectable (**Fig. 4** [↗](#)).

## More accurate sampling of simulated neuronal activity with long corrected microendoscopes

Using simulated calcium t-series, we next evaluated the impact of the improved optical performance of long aberration corrected microendoscopes on the ability to extract information about neuronal activity. To this aim, we compared simulated calcium traces imaged with either uncorrected or corrected microendoscopes with the ground truth neuronal activity, i.e. the neuronal activity used to generate the simulated t-series. To build synthetic calcium data, we first generated neurons with 3D distribution and anatomical properties (i.e., cell size and cell density) similar to those measured in the mouse olfactory cortex [29](#) [30](#), the ventral brain region which we will be imaging *in vivo* (see next section of Results), and the biophysical characteristics of one of the two indicators used in our experiments, jRCaMP1f [31](#) (see also Materials and Methods). We then simulated the optical sampling of 3D volume of neurons with the PSF properties of the uncorrected and corrected microendoscopes characterized above (**Fig. 3** [↗](#)). Fluorescence signals integrated over the PSF volume were then projected on a 2D matrix of pixels in time, with a frequency of 30 Hz, to obtain synthetic t-series (duration: 5 min, **Fig. 5A, D** [↗](#)). We finally detected neuronal cell bodies and extracted fluorescence traces from regions of interest (ROIs) of synthetic t-series using established methods [32](#) commonly applied to real data and compared between the case of uncorrected and corrected microendoscopes. We found that the use of corrected probes allowed segmenting more ROIs with high SNR compared to uncorrected microendoscopes, shifting the distribution of SNR across ROIs to higher mean SNR values. This was true for both the microendoscopes based on the 6.4 mm GRIN rod (**Fig. 5B** [↗](#)) and the microendoscope based on the 8.8 mm long GRIN rod (**Fig. 5E** [↗](#)). Moreover, for both microendoscope types, aberration correction increased the maximal radial distance at which ROIs with high ( $> 15$ ) SNR could be observed (**Fig. 5C, F** [↗](#)). These findings demonstrate that correcting for optical aberrations increases the probability of recording ROIs with larger SNR of fluorescent signals.

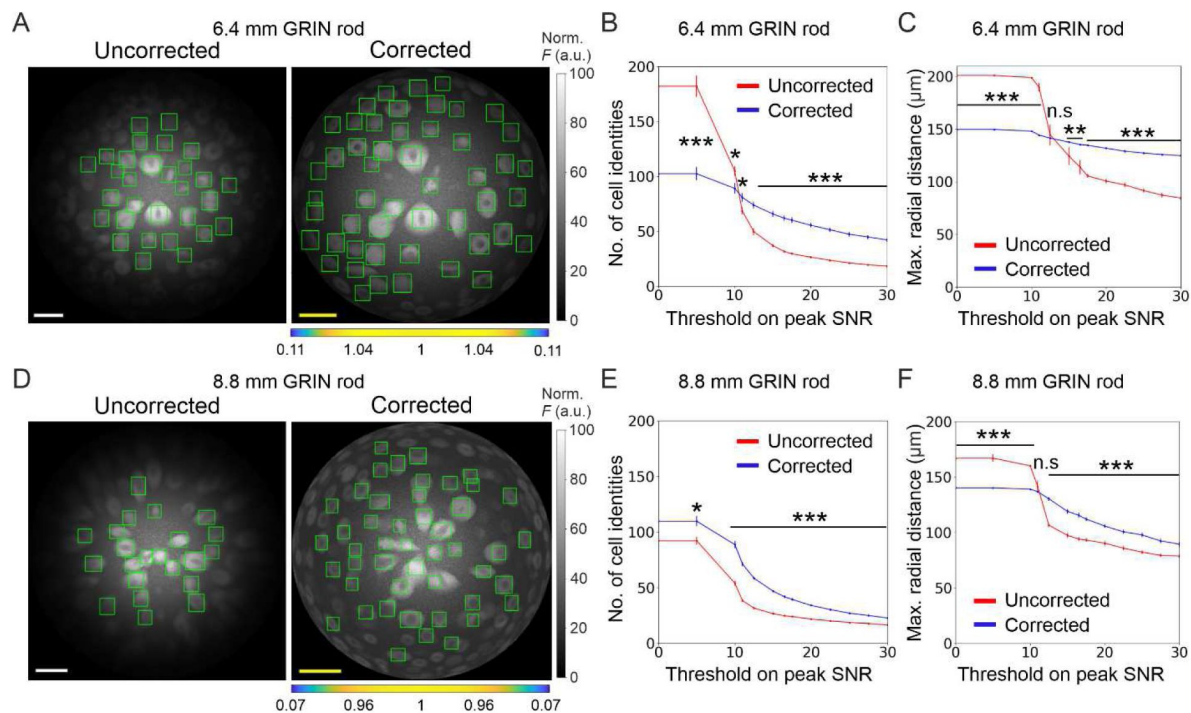
The higher and more homogeneous axial resolution across the FOV afforded by the corrected probes could be useful not only to increase the SNR of neural calcium signals, but also to better isolate individual neurons, decreasing signal contamination across neighboring cells. To test this hypothesis, we considered correlations in “adjacent” simulated neurons, defined as pairs of neurons for which the distance between their centroids was  $\leq 25 \mu\text{m}$  (the soma of excitatory neurons in the olfactory cortex is around  $10 \mu\text{m}$  [30](#)). Compared to uncorrected probes, we found that corrected microendoscopes of both lengths led to lower fraction of pairs of adjacent cells whose activity correlated significantly more than expected (**Fig. 6A, F** [↗](#) and **Supplementary Table 5** [↗](#)), confirming reduced cross contamination among cells. The fraction of pairs of adjacent cells (out of the total number of adjacent pairs) whose activity correlated significantly more than expected increased as a function of the SNR threshold for corrected and uncorrected microendoscopes of both lengths (**Fig. 6A, F** [↗](#)). This effect was due to a larger decrease of the total number of pairs of adjacent cells as a function of the SNR threshold compared to the decrease in the number of pairs of adjacent cells whose activity was more correlated than expected (**Supplementary Figure 5** [↗](#)). Moreover, in uncorrected microendoscopes the pairwise correlation of adjacent cells showed a larger linear dependence on the radial distance of the pair from the center of the FOV compared to corrected microendoscopes (**Supplementary Figure 6A, C** [↗](#), see also Materials and Methods). We interpreted these findings on correlated neuronal activity as due to the enlarged PSF of uncorrected microendoscopes in lateral portion of the FOV (**Fig. 3** [↗](#)), leading to increased signal contamination by sampling across neurons. These results thus suggest



**Figure 4.**

**Aberration correction in long GRIN lens-based microendoscopes enables high-resolution imaging of biological structures over enlarged FOVs.**

A) jRCaMP7f-stained neurons in a fixed mouse brain slice were imaged using 2PLSM ( $\lambda_{\text{exc}} = 920 \text{ nm}$ ) through an uncorrected (left) and a corrected (right) microendoscope based on the 6.4 mm-long GRIN rod. Images are maximum fluorescence intensity ( $F$ ) projections of a z-stack acquired with a  $5 \mu\text{m}$  step size. Number of steps: 32 and 29 for uncorrected and corrected microendoscope, respectively. Scale bars:  $50 \mu\text{m}$ . Left: the scale applies to the entire FOV. Right, the scale bar refers only to the center of the FOV; off-axis scale bar at any radial distance ( $x$  and  $y$  axes) is locally determined multiplying the length of the drawn scale bar on-axis by the corresponding normalized magnification factor shown in the horizontal color-coded bar placed below the image (see also [Fig. 3](#), [Supplementary Table 3](#), and Materials and Methods for more details). B) Same results for the microendoscope based on the 8.8 mm-long GRIN rod. Number of steps: 23 and 31 for uncorrected and corrected microendoscope, respectively.



**Figure 5.**

### Long corrected microendoscopes sample more homogeneously simulated neuronal activity across the FOV.

A) Median fluorescence intensity ( $F$ ) projections of representative synthetic t-series for the uncorrected (left) and the corrected (right) 6.4 mm-long microendoscope. Cell identities were detected using CITE-ON [32](https://doi.org/10.7554/eLife.101420.2) in  $n = 13$  simulated t-series for both the uncorrected and corrected microendoscopes and cellular activity traces were extracted. Green rectangular boxes mark cell identities that have peak SNR of the activity trace higher than a threshold set to peak SNR = 15. Scale bars: 50  $\mu\text{m}$ . Left: the scale applies to the entire FOV. Right: the scale bar refers to the center of the FOV; off-axis scale bar at any radial distance ( $x$  and  $y$  axes) is locally determined multiplying the length of the drawn scale bar by the corresponding normalized magnification factor shown in the horizontal color-coded bar placed below the image. B) Number of detected cell identities in simulated FOV as a function of peak SNR threshold imposed on cellular activity traces. Data are mean values  $\pm$  s.e.m. for both the uncorrected (red) or corrected (blue) case. Statistical significance is assessed with Mann-Whitney U test; \*,  $p < 0.05$ ; \*\*\*,  $p < 0.001$ . C) Maximal distance from the center of the FOV at which a cell is detected as a function of peak SNR threshold. Data are mean values  $\pm$  s.e.m. for both the uncorrected (red) and corrected (blue) case. Statistical significance is assessed with Mann-Whitney U test; \*\*,  $p < 0.01$ ; \*\*\*,  $p < 0.001$ ; n.s., non significant. D) Same as (A) for the 8.8 mm-long microendoscope. E, F) Same as (B,C) for  $n = 15$  simulated t-series for both the uncorrected and corrected 8.8 mm-long microendoscope.

that corrected microendoscopes decrease the artefactual correlation of nearby neurons by diminishing the spatial extension of the excitation volume used for sampling neuronal activity (PSF, [Fig. 3](#) [↗](#)).

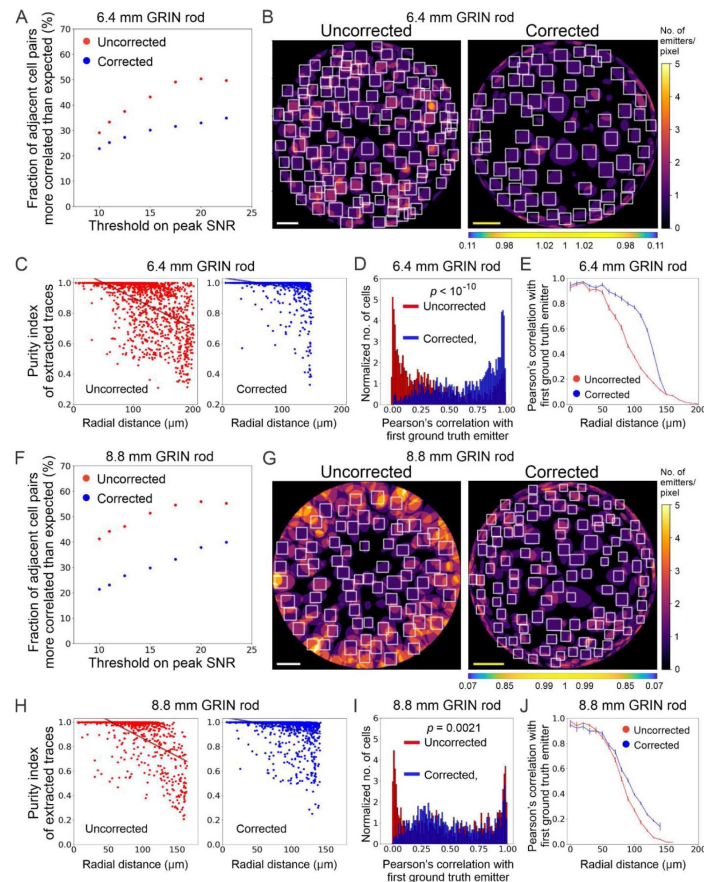
Using synthetic data, we could precisely quantify the extent to which correcting aberration in microendoscopes decreased mixing signals across neurons. For each ROI, we computed a general linear model (GLM) of the contributions to fluorescence signals of the activity of all ground truth simulated neurons, called “sources”, [Fig. 6B, C, G, H](#) [↗](#)). From the GLM coefficients, we extracted a source “purity” index, which reached its highest possible value of 1 when only one source contributed to the calcium fluorescent trace in the considered ROI and had values < 1 when different source neurons were mixed in the fluorescent signal of the considered ROI. For uncorrected microendoscopes of both lengths, we found that purity rapidly decreased with the radial distance of the ROI from the center of the FOV. In contrast, for corrected microendoscopes the purity index decreased far more slowly with the radial distance and the slope of the linear fit of the purity distribution as a function of the radial distance was significantly smaller than that of the uncorrected probe ([Fig. 6C, H](#) [↗](#)). For any value of the SNR threshold used to select neurons, the purity index was also larger for corrected than for uncorrected probes ([Supplementary Figure 6A, C](#) [↗](#)). Moreover, for each ROI we analyzed the correlation of the extracted traces with the most correlated ground truth neuron (“first ground truth source”) as a function of the radial distance of the ROI ([Fig. 6D, E, I, J](#) [↗](#)). For microendoscopes of both lengths, we observed that ROIs from t-series generated with corrected microendoscopes displayed larger correlation with the ground truth signal of an individual neuron over longer distances from the optical axis, compared to uncorrected probes ([Fig. 6E, J](#) [↗](#)). [Figure 6J](#) [↗](#) showed a moderate enlargement of the distance at which the Pearson’s correlation with the first ground truth emitter started to drop. This may appear at first in contrast with the enlargement of the FOV shown in [Figure 4B](#) [↗](#). However, it can be understood considering that in [Figure 4B](#) [↗](#), all illuminated neurons were visible regardless of whether they were imaged with high axial resolution (e.g., < 10  $\mu\text{m}$  as defined in [Figure 3J](#) [↗](#)) or poor axial resolution. In contrast, in [Figure 6J](#) [↗](#) we evaluated the correlation between the calcium trace extracted from a given ROI and the real activity trace of the first simulated ground truth emitter for that specific ROI. The moderate increase in the correlation for the corrected microendoscope compared to the uncorrected microendoscope ([Figure 6J](#) [↗](#)) was consistent with the moderate improvement in the axial resolution of the corrected probe compared to the uncorrected probe at intermediate radial distances (60–100  $\mu\text{m}$  from the optical axis, see [Figure 3J](#) [↗](#)). Taken together, these results indicated that corrected microendoscopes enabled more precise recording of individual cellular source neurons and enhanced demixing of signals coming from different neurons.

## Validation of long corrected microendoscopes for large FOV population imaging in ventral brain region of awake mice

We tested long corrected microendoscopes for *in vivo* 2P calcium imaging experiments in ventral regions of the mouse brain. To this aim, we implanted mice with corrected microendoscopes based on 8.8 mm-long GRIN lenses over the mouse anterior olfactory cortex, a deep region laying > 4 mm down the mouse brain surface ([Fig. 7A, B](#) [↗](#)). Mice expressed genetically encoded calcium indicators, either jRCaMP7f or jRCaMP8f, in excitatory neurons through adeno-associated virus (AAVs) injection. In awake head-fixed mice, we recorded 2P imaging t-series ( $\lambda_{\text{exc}} = 920 \text{ nm}$ ; frame rate: 30 Hz; t-series duration: 8 minutes) during repetitive olfactory stimulation with various odorants ([Fig. 7C–E](#) [↗](#), see Materials and Methods).

In recorded t-series, we first observed that the SNR of calcium events recorded in individual cells was constant across the FOV ([Fig. 8A](#) [↗](#)). We then computed the pairwise correlation of the activity traces of pairs of adjacent neurons (defined as pairs of cells with an intersomatic distance of < 25  $\mu\text{m}$ ) as a function of the radial distance of the pair’s position centroid from the center of the FOV ([Fig. 8B](#) [↗](#)). We found no dependence of the pairwise correlation on the distance from the center

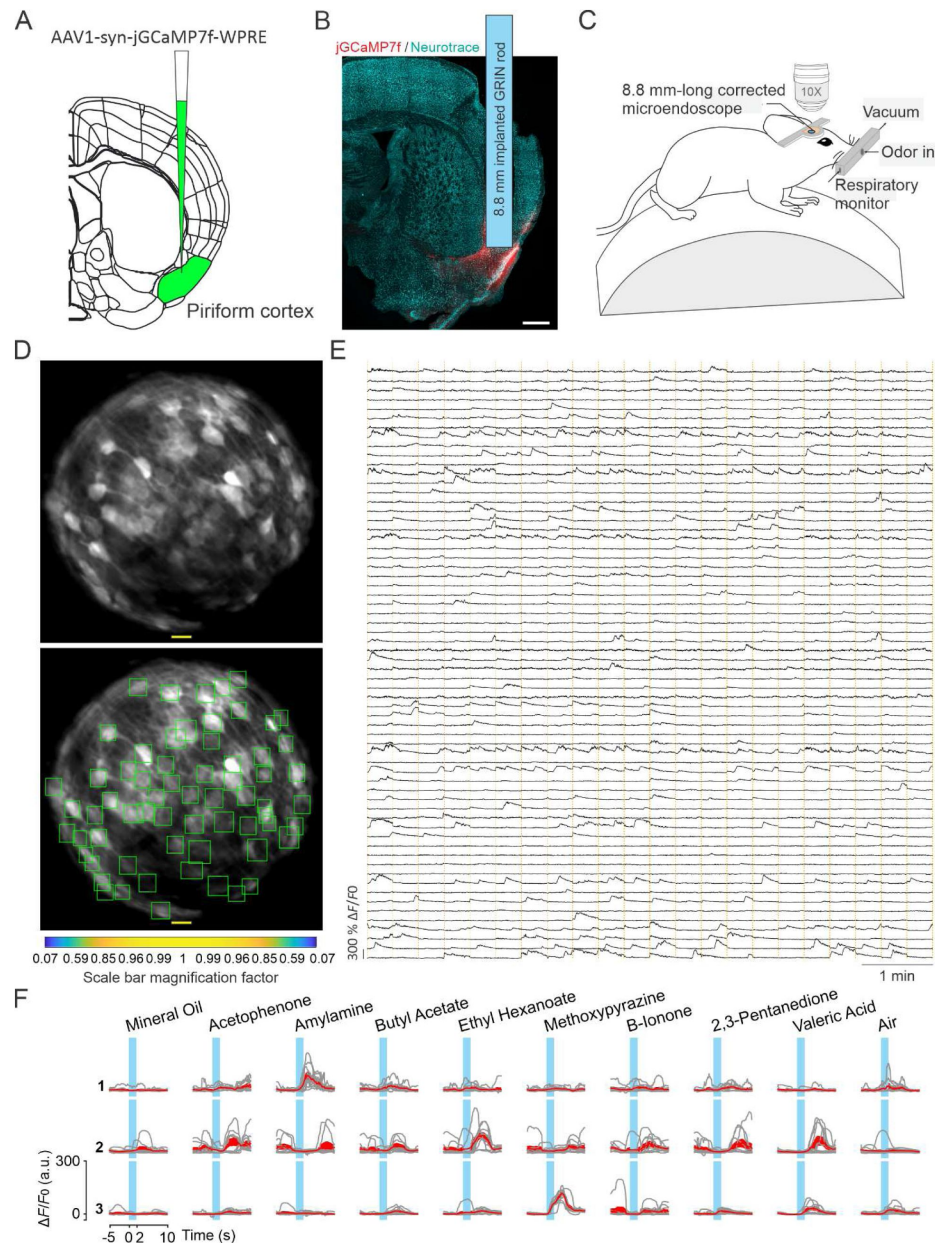




**Figure 6.**

**Long corrected microendoscopes enable more precise collection of simulated activity signals from individual cellular sources and decrease cross-contamination between adjacent cells.**

A) Fraction of adjacent cell pairs (distance between detected cell centroids  $\leq 25 \mu\text{m}$ ) that are more correlated than expected (expected pair correlation was estimated as mean Pearson's correlation between ground truth activity traces of any possible neuronal pairs plus 3 standard deviations) as a function of the peak SNR threshold imposed on extracted activity traces for  $n = 13$  simulated experiments with the 6.4 mm-long uncorrected (red) and corrected (blue) microendoscope. B) 2D projection of the intersection between the 3D FOV and the 3D ground truth distribution of light sources for a representative uncorrected (left) and corrected (right) synthetic t-series obtained with a 6.4 mm-long microendoscope. The color scale shows the number of overlapping sources that are projected on the same pixel. White boxes mark cell identities detected using CITE-ON [32](#) without any threshold on the peak SNR of activity traces. Scale bars:  $50 \mu\text{m}$ . Left: the scale applies to the entire FOV. Right: the scale bar refers to the center of the FOV; off-axis scale bar at any radial distance ( $x$  and  $y$  axes) is locally determined multiplying the length of the scale bar by the corresponding normalized magnification factor shown in the horizontal color-coded bar placed below the image. C) Purity index of extracted traces with peak SNR  $> 10$  was estimated using a GLM of ground truth source contributions and plotted as a function of the radial distance of cell identities from the center of the FOV for  $n = 13$  simulated experiments with the 6.4 mm-long uncorrected (red) and corrected (blue) microendoscope. Black lines represent the linear regression of data  $\pm 95\%$  confidence intervals (shaded colored areas). Slopes  $\pm$  standard error (s.e.): uncorrected,  $(-0.0020 \pm 0.0002) \mu\text{m}^{-1}$ ; corrected,  $(-0.0006 \pm 0.0001) \mu\text{m}^{-1}$ . Uncorrected,  $n = 1365$ ; corrected,  $n = 1156$ . Statistical comparison of slopes,  $p < 10^{-10}$ , permutation test. D) Distribution of the Pearson's correlation of extracted activity traces with the first (most correlated) ground truth source for  $n = 13$  simulated experiments with the 6.4 mm-long uncorrected (red) and corrected (blue) microendoscope. Median values: uncorrected, 0.25; corrected, 0.73; the  $p$  is computed using the Mann-Whitney U test. E) Pearson's correlation  $\pm$  s.e.m. of extracted activity traces with the first (most correlated) ground truth emitter as a function of the radial distance for  $n = 13$  simulated experiments with the 6.4 mm-long uncorrected (red) and corrected (blue) microendoscope. F-J) Same as (A-E) for  $n = 15$  simulated experiments with the 8.8 mm-long uncorrected and corrected microendoscope. H) Slopes  $\pm$  s.e.: uncorrected,  $(-0.0031 \pm 0.0003) \mu\text{m}^{-1}$ ; corrected,  $(-0.0010 \pm 0.0002) \mu\text{m}^{-1}$ . Uncorrected,  $n = 808$ ; corrected,  $n = 1328$ . Statistical comparison of slopes,  $p < 10^{-10}$ , permutation test. I) Median values: uncorrected, 0.43; corrected, 0.46; the  $p$  is computed using the Mann-Whitney U test.



**Figure 7.**

### Enlarged FOV population imaging of ventral regions of the brain with long corrected microendoscopes in awake mice.

A) Schematic showing injection of the viral solution in the mouse piriform cortex. B) Representative section of fixed brain tissue showing the position of the GRIN lens implant. jGCaMP7f fluorescence is shown in red. NeuroTrace (NeuroTrace) Nissl staining is shown in cyan. C) Schematic showing the experimental preparation for 2P corrected microendoscope imaging in awake mice. D) Top and bottom: example of FOV with excitatory neurons expressing the calcium sensor jGCaMP7f, obtained using the 8.8 mm-long corrected microendoscope in the mouse PC. The bottom image shows the rectangular boxes (green) indicating the position of detected neurons generated by CITE-ON (see Materials and Methods, [32](#)). On-axis scale bar: 20  $\mu\text{m}$ ; for off-axis scale bar, refer to the magnification factor bar below the image (see Materials and Methods). E) Representative jGCaMP7f activity traces extracted from the recording shown in (D) through eight minute-long continuous 2P imaging recordings ( $\lambda_{\text{exc}} = 920 \text{ nm}$ ) in an awake head-fixed mouse. Vertical dashed orange lines mark the onset and the end of the 22 s-long trials in which the imaging session was divided. F) Examples of neuronal responses to different olfactory stimuli for three representative cells. For each odor (top labels), gray lines represent calcium responses recorded in single trials and the red line represent the average response. Light blue areas indicate the time interval of stimulus presentation.

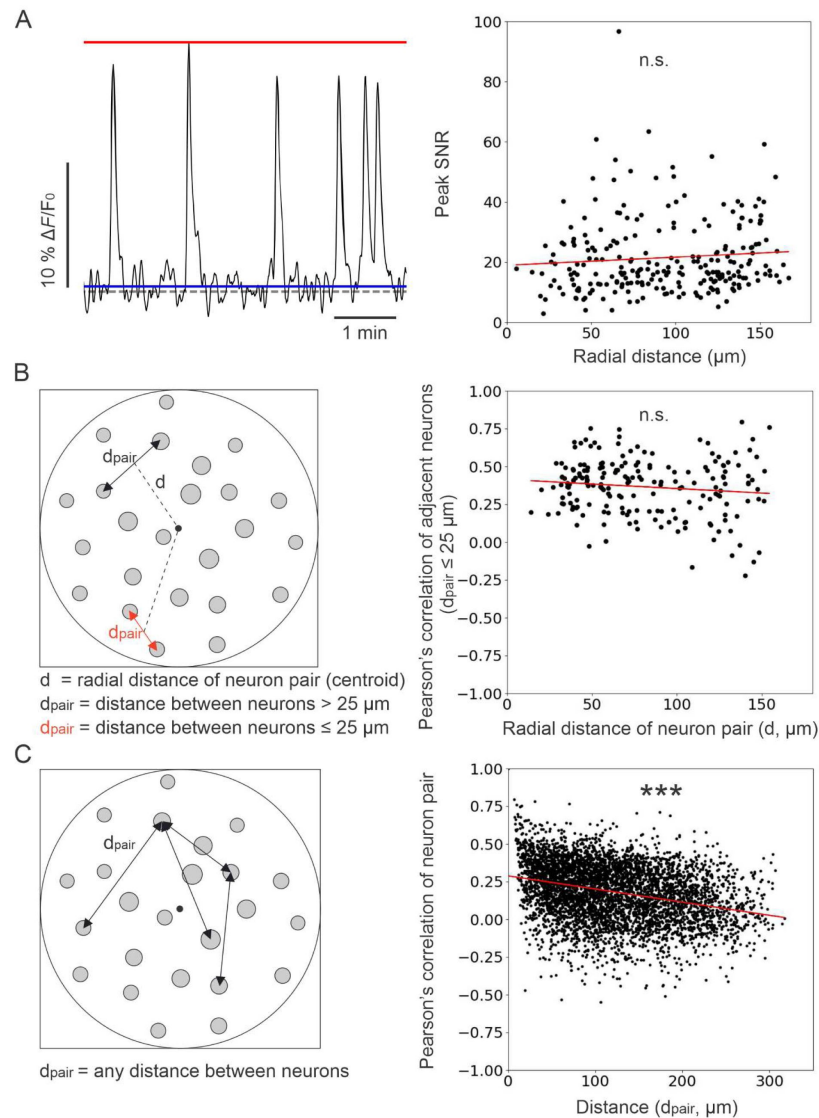
of the FOV. In fact, the slope of the linear fit of pairwise correlation as a function of the radial distance was not significantly different from zero (**Fig. 8B**). These findings were in agreement with the results of the optical characterization (**Fig. 3**) and with those of the simulation of calcium data (**Fig. 5**, 6). These results suggested that the absence of dependence of the pairwise correlation with radial distance reflected the improved and more homogeneous spatial resolution of corrected microendoscopes, which prevented source mixing and artificially inflated correlations of adjacent neurons in the FOV periphery. Varying the distance between centroids of adjacent neurons did not change the result (**Supplementary Table 6**). Running the analysis separately for mice expressing jGCaMP8f and jGCaMP7f also confirmed the previous findings (**Supplementary Table 6**). Finally, the Pearson's correlation coefficient between the activity traces of any pair of neurons within the FOV decreased as a function of their reciprocal distance (**Fig. 8C**). This decrease in correlation could reflect a slight change in the laminar position of recorded neurons or suggest a decrease in recurrent connectivity between neurons of the olfactory cortex with distance<sup>33, 34</sup>.

Combined with the results of the simulated calcium data, these *in vivo* findings demonstrate that, by decreasing off-axis aberrations of commercial GRIN lenses, long corrected microendoscopes enable unbiased sampling of neural population activity on a large FOV in ventral regions of the mouse brain *in vivo*.

## Discussion

Combining GRIN lens-based endoscopy<sup>8, 9</sup> with 2P fluorescence imaging<sup>3, 35</sup> is a powerful approach to study the physiology of deep brain circuits. However, GRIN lenses are not ideal optical elements and they have intrinsic optical aberrations<sup>18, 19</sup>, which significantly degrade the PSF and reduce the effective FOV<sup>7, 20, 21, 23</sup>. In the current study, we developed, characterized, and validated new aberration corrected microendoscopic probes to address this critical need. Corrected microendoscopes had length of 6.4 mm and 8.8 mm (see **Table 2** and Materials and Methods), allowing reaching the deepest structures of the mouse brain, and cross section of 500  $\mu\text{m}$ , ensuring limited tissue invasiveness.

We first designed the corrected microendoscopes using optical ray-trace simulations. Simulations showed strongly aberrated PSF in lateral portions of the FOV of uncorrected microendoscopes and predicted that single corrective optical elements with the aspherical profiles shown in **Fig. 1A-B** improved the optical performance of microendoscopes of either length. This prediction was experimentally confirmed using subresolved fluorescent beads and layers. We first observed that uncorrected microendoscopes of either length displayed strong astigmatism and had largely distorted PSF in lateral regions of the FOV. In corrected microendoscopes, we instead found that the PSF had regular shape and constrained values for larger radial distances, leading to more homogeneous spatial resolution across the FOV and enlarged effective FOV. Importantly, experimental values describing the improvement of the optical performance of the corrected microendoscopes (**Fig. 3G-J** and **Supplementary Table 4**) were generally in good agreement with the prediction of the optical simulations (**Fig. 1D, F** and **Supplementary Table 2**). The size of simulated PSFs at a given radial distance (e.g., 90  $\mu\text{m}$ , **Figure 1**) tended to be generally smaller than that of the experimentally measured PSFs (**Figure 3**). This might be due to multiple reasons. First, simulated PSFs were excitation PSFs, i.e. they described the intensity spatial distribution of focused excitation light. On the contrary, measured PSFs resulted from the excitation and emission processes and thus they were also affected by aberrations of light emitted by fluorescent beads and collected by the microscope. Second, in the optical simulations first-order and not higher-order aberrations were considered. Third, intrinsic variability in experimental parameters (e.g., alignment of the microendoscope to the optical axis of the microscope, distance between the GRIN back end and the objective) were not considered in the simulations.



**Figure 8.**

### Unbiased population imaging in the piriform cortex of awake mice using long corrected microendoscopes.

A) Left: schematic representation of trace parameters used to compute peak SNR. The red solid line marks the maximum value of the trace, the blue line indicates the standard deviation of intensity values below the 25<sup>th</sup> percentile of the intensity distribution of the entire trace (see Materials and Methods), and the dashed grey line marks the origin of the y-axis ( $\Delta F/F_0 = 0$ ). Right: peak SNR of calcium traces extracted from individual cells as a function of radial distance of the cell ( $n = 240$  neurons from  $m = 6$  FOV). The red line is the linear regression of data points: intercept  $\pm$  s.e. =  $19 \pm 2$ ; slope  $\pm$  s.e. =  $(0.03 \pm 0.02) \mu\text{m}^{-1}$ . The slope is not significantly different from zero,  $p = 0.18$ , permutation test. B) Left: schematic representation showing pairs of neurons and distance definitions for the Pearson's correlation analysis shown on the right (in red, distance between "adjacent neurons"  $d_{\text{pair}} \leq 25 \mu\text{m}$ ). Right: Pearson's correlation of calcium traces from "adjacent neurons" as a function of radial distance of the pair centroid from the center of the FOV ( $n = 195$  adjacent neuron pairs from  $m = 6$  FOV). The red line represents the linear regression of data points: intercept  $\pm$  s.e. =  $0.41 \pm 0.03$ ; slope  $\pm$  s.e. =  $(-0.0006 \pm 0.0004) \mu\text{m}^{-1}$ . The slope is not significantly different from zero,  $p = 0.089$ , Wald test. C) Left: schematic representation of neuron pairs used for the analysis on the right (any possible  $d_{\text{pair}}$ ). Right: Pearson's correlation of calcium traces from pairs of neurons as a function of the distance between them ( $n = 4767$  pairs from  $m = 6$  FOV). The red line is the linear regression of data points: intercept  $\pm$  s.e. =  $0.288 \pm 0.005$ ; slope  $\pm$  s.e. =  $(-87 \pm 4) \cdot 10^{-5} \mu\text{m}^{-1}$ . The slope is significantly different from zero,  $p = 2 \cdot 10^{-5}$ , permutation test.

|                              | 6.4 mm-long GRIN rod                |                                     | 8.8 mm-long GRIN rod                |                                     |
|------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
|                              | Commercial                          | Custom                              | Commercial                          | Custom                              |
| <b>Catalogue #</b>           | NEM-050-25-10-860-S-1.5p            | NEM-050-25-15-860-S-1.5p            | NEM-050-25-10-860-S-2.0p            | NEM-050-23-15-860-S-2.0p            |
| <b>Diameter (μm)</b>         | 500                                 | 500                                 | 500                                 | 500                                 |
| <b>Length (mm)</b>           | 6.52                                | 6.4                                 | 8.85                                | 8.8                                 |
| <b>Working distance (μm)</b> | Image side: 100<br>Object side: 250 | Image side: 150<br>Object side: 250 | Image side: 100<br>Object side: 250 | Image side: 150<br>Object side: 230 |

**Table 2.**

**Characteristics of commercially available (commercial) and customized (custom) GRIN rods used for simulation and fabrication of aberration corrected microendoscopes.**

All GRIN rods were obtained from GRINTECH GmbH, Jena, DE.



Did the improved optical properties of corrected microendoscopes increase the quality of imaging when corrected microendoscopes were applied to biological samples? We directly addressed this question by performing 2P imaging with uncorrected and corrected microendoscopes in brain slices in which neurons expressed a green fluorescent indicator. In uncorrected microendoscopes, we found that neuronal cell bodies were bright and well contrasted only in the center of the FOV and that neuronal cell bodies were dimmer and blurred in lateral regions of the FOV (**Fig. 4** [↗](#)). Moreover, neuronal processes could be observed only in the center of the FOV and not in lateral portions of the FOV (**Fig. 4** [↗](#)). In contrast, in corrected microendoscopes of either length, we found that neurons were bright and well contrasted across the whole FOV, which enabled distinguishing neurons, which were blurred and merged together when imaged with uncorrected microendoscopes (**Fig. 4** [↗](#)). Furthermore, small neuronal processes could be observed throughout the entire FOV. These imaging findings in brain slices are in line with the results of the optical simulation and of the experimental characterization, which showed improved PSF and more homogeneous spatial resolution in corrected microendoscopes compared to uncorrected ones. It must be considered, however, that the extended FOV achieved by our aberration correction method was characterized by a curved focal plane. Therefore, cells located in different radial positions within the image were located at different axial positions and cells at the border of the FOV were closer to the front end of the microendoscope.

How does the improved optical performance of corrected microendoscope translate into enhanced ability to extract meaningful information from population of neurons in functional fluorescence imaging experiments? We addressed this question generating and analyzing synthetic calcium data in which we could compare the signals extracted from simulated recordings using uncorrected and corrected microendoscopes with the ground truth signals, which were used to generate the simulations. We found that the aberration correction led to an increase in the number of ROI with high SNR and, correspondingly, a decrease in the number of ROIs with low SNR. Moreover, aberration correction led to high and more stable purity of source demixing over radial distance, which, in turn, led to a major decrease in the artificial overestimation of correlations between adjacent neurons due to source mixing. Since investigating neuronal population codes needs quantification of neuronal activity with large SNR, high precision, and limited contamination, we expect the corrected microendoscopes presented in this manuscript will greatly contribute to the investigation of deep brain regions. In particular, the reduction of artificial overestimation of correlations due to source mixing appears crucial to precisely evaluate within-network neuron-to-neuron communication and investigate the role of correlations between neurons in population codes. Artificial correlations between nearby neurons would bias estimates of neuron-to-neuron communication toward overestimating communications at short distances. Moreover, because source mixing duplicates one neuron into another and thus can only increase redundancy, artificial correlations would also bias the effect of correlations on population coding by artificially inflating redundant correlations over synergistic interactions between neurons <sup>36</sup>[↗](#), <sup>37</sup>[↗](#). Thus, correcting optical aberrations in GRIN lenses as presented in this study appears essential for the unbiased characterization of single-neuron and population codes in deeper brain structures.

We used long corrected microendoscopes to measure population dynamics in the olfactory cortex of awake head-restrained mice with unprecedented combination of high spatial resolution across the FOV and minimal invasiveness <sup>17</sup>[↗](#). In agreement with the results of our optical characterization and of the synthetic calcium data, we found that the SNR of calcium signals was independent on the radial distance and that high SNR calcium signals could be recorded across the whole FOV. Moreover, we observed that the correlation between calcium signals of adjacent neurons did not depend on the radial distance, in line with the more homogeneous PSF that characterizes corrected microendoscopes. The correlation between the calcium signals of any pair of neurons displayed, instead, a negative dependence on the distance between the two neurons. This decrease in correlation could reflect a slight change in the laminar position of recorded neurons or suggest a decrease in recurrent connectivity between neurons of the olfactory cortex

with distance <sup>33</sup>, <sup>34</sup>. In relationship to studies on the olfactory cortex, the improved optical characteristics of corrected microendoscopes increase our ability to understand the properties of this network in several fundamental ways. First, imaging more cells with increased SNR improves our ability to decipher information encoded by neural ensembles in the olfactory cortex. For example, we will be able to characterize the odor response properties of genetically defined subpopulations of olfactory cortex neurons, which may be too sparse to be reliably captured in the smaller effective FOV of uncorrected GRIN lenses. Second, imaging more cells with high SNR in each mouse will allow us to explore differences between individual mice, rather than pooling cells from different animals. Third, less cross-contamination of fluorescence signals improves cell segmentation and motion correction, which are major challenges when imaging mice that are awake or actively engaged in behavioral tasks <sup>38</sup>, <sup>39</sup>. Finally, improved cell segmentation facilitates cell tracking in chronic imaging experiments. Thus, corrected microendoscopes will enhance our ability to monitor individual cells over extended periods of time, for example while mice learn to perform an odor discrimination task <sup>40</sup>.

Alternative methods to correct optical aberrations in GRIN lenses include, for example, low-order adaptive optics <sup>18</sup>, adaptive optics through pupil segmentation <sup>20</sup>, <sup>21</sup>, and geometric transformation adaptive optics <sup>23</sup>. In the first approach, an adaptive optics module comprising an electrostatic membrane mirror was introduced in the optical path of the 2P microscope and it was shown to significantly improve the microscope PSF <sup>18</sup>. In adaptive optics through pupil segmentation instead, the objective rear lens was divided into N different subsections. By comparing images acquired with light going through each individual pupil segment with images acquired with light going through the whole pupil, local wavefront tilts introduced by aberrations were calculated and corrected for using a spatial light modulator (SLM) <sup>41</sup>. This method was applied to GRIN lens-based endoscopes of larger diameter (i.e., 1.4 mm) compared to the one used in this study (i.e., 0.5 mm) and it was shown to enlarge the effective FOV during 2P imaging <sup>20</sup>, <sup>21</sup>. Finally, in geometric transformation adaptive optics, a 90-degrees rotation was applied to the aberrated beam spatial profile exiting a first GRIN lens and the aberrated beam was passed through a second GRIN lens. This strategy was effective in correcting chromatic and volumetric aberrations in GRIN microendoscopes of small cross section (500  $\mu\text{m}$  <sup>23</sup>). However, all adaptive optics methods described above require substantial change the optical set-up (e.g., the insertion of an SLM, membrane mirror, and additional optics) and the development of specific software control <sup>18</sup>, <sup>20</sup>, <sup>21</sup>, <sup>23</sup>. Additionally, under certain conditions pupil segmentation may limit the acquisition frame rate of the system <sup>21</sup>. In contrast to the methods described above, the long corrected microendoscopes presented in the current manuscript do not limit the temporal resolution of the acquisition system, require no modification of the hardware and software of the optical set-up, and are easily coupled to any standard 2P microscope. Another advantage of long corrected microendoscopes described here over adaptive optics approaches is the possibility to couple corrected microendoscopes with portable 2P microscopes <sup>42–44</sup>, allowing high resolution functional imaging of deep brain circuits on an enlarged FOV during naturalistic behavior in freely moving mice. The corrected microendoscopes presented in this study thus represent a ready-to-use solution, which will likely ease the adoption of aberration correction in GRIN lens-based 2P functional imaging across experimental laboratories.

In conclusion, we designed, developed, and validated a new series of aberration-corrected GRIN lens-based microendoscopes, which are long enough to reach the most ventral regions of the mouse brain. Long corrected microendoscopes had improved axial resolution and several folds enlarged effective FOV. Importantly, these optical improvements enabled more precise extraction of single cell and population signals from 2P imaging recordings. We validated long corrected microendoscopes by performing functional imaging of the olfactory cortex circuits in awake head-fixed mice. Long corrected microendoscope provide a unique combination of high and homogenous optical properties across the FOV and minimal invasiveness. Furthermore, they do not require modification of the hardware and software of the optical microscope, providing a ready-to-use solution to experimental laboratories. We foresee that the long corrected

microendoscopes developed in this study will drastically improve the yield of 2P imaging in deep brain areas of mice and larger mammals, such as rats <sup>45</sup>, cats <sup>46</sup>, <sup>47</sup>, and marmosets <sup>48</sup>, <sup>49</sup>.

## Materials and Methods

### Ray-trace simulations of corrected microendoscopes

To identify the design of the corrective optics to obtain aberration corrected microendoscopes, we performed ray-trace simulations using OpticStudio 15 (Zemax, Kirkland, WA, US) and integrating the optical model of two commercially available GRIN rods (NEM-050-25-10-860-S-1.5p and NEM-050-25-10-860-S-2.0p, GRIN needle endomicroscopes, GRINTECH GmbH, Jena, Germany). The wavelength selected for simulations was 920 nm, the commonly used excitation wavelength for 2P imaging of green-emitting fluorescent indicators <sup>50</sup>, <sup>51</sup>. In simulations, we placed the aspheric lens, whose refractive index corresponded to that of the UV-curable glue we used for microendoscope fabrication (NOA63, see also below), on one side and the back end of the GRIN rod on the other side. 100  $\mu\text{m}$  of BK7 glass were interposed between the corrective lens and the GRIN rod. The profile of the aspheric lens was identified by optimizing a merit function, which took into account the Strehl ratio of focal spots simulated at different field radial distances (up to 200  $\mu\text{m}$  for the 6.4 mm-long and up to 170  $\mu\text{m}$  for the 8.8 mm-long GRIN rod) after being relayed by the optical assembly. During the optimization, the working distance was constrained to values compatible with neurophysiological experiments ( $\sim 180$ -220  $\mu\text{m}$ ) and the refractive indices simulated at the two ends of the optical assembly were 1 (air) and 1.33-1.36 (brain tissue). The following formula described the surface profile of aspheric lenses similarly to <sup>7</sup>:

$$Z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_n \alpha_n r^{2n} \quad (1)$$

We changed the parameters  $c$ ,  $k$ ,  $\alpha_n$  (with  $n = 1$ -8) in equation (1) to maximize the Strehl ratio <sup>25</sup> over the largest possible area of the FOV. We obtained simulated 2P PSFs (Fig. 1) by 3D sampling the squared calculated Strehl ratio <sup>25</sup>. To compute the spatial resolution of simulated microendoscopes, we fitted x,y,z intensity profiles of the simulated PSFs with Gaussian curves. The ray-trace simulations demonstrated improved Strehl ratio values with GRIN lenses of slightly longer working distances (Table 2, Custom) compared to the two commercial GRIN rods initially considered (Table 2, Commercial). We thus used the GRIN lenses of custom working distance indicated in Table 2 for all results presented in this manuscript. To evaluate the impact of corrective microlenses on the optical performance of GRIN-based microendoscopes, we also simulated uncorrected microendoscopes composed of the same optical elements of corrected probes (glass coverslip and GRIN rod), but in the absence of the corrective microlens. Final working distances of simulated uncorrected and corrected microendoscopes based on custom GRIN rods are indicated in Table 3.

### Aspherical lens microprinting and microendoscope assembly

The aspheric profiles of the corrective lenses obtained with optical simulations were converted into 3D structure files by a CAD software (SolidWorks, Waltham, MA). The aspherical lens prototypes were then fabricated with a commercial 3D microprinter based on 2P polymerization (Photonic Professional GT2+, Nanoscribe, Karlsruhe, DE) using a proprietary photoresin (IP-S, Nanoscribe, Karlsruhe, DE) and a 25x 0.8-NA microscope objective in Dip-in Laser Lithography (DiLL) configuration. Once the lens' volume was fully polymerized in Solid printing mode, the residual unpolymerized material was removed with immersion in propylene glycol methyl ether acetate (PGMEA) for 10 minutes, followed by  $\sim 5$ -minute immersion in isopropyl alcohol. The lens prototypes were finally coated with 200 nm of gold with a sputter coater and inspected with scanning electron microscopy to confirm absence of surface defects.

|                                               | Microendoscope based on 6.4 mm-long GRIN rod |                                     | Microendoscope based on 8.8 mm-long GRIN rod |                                     |
|-----------------------------------------------|----------------------------------------------|-------------------------------------|----------------------------------------------|-------------------------------------|
|                                               | Uncorrected                                  | Corrected                           | Uncorrected                                  | Corrected                           |
| <b>Working distances</b><br>( $\mu\text{m}$ ) | Image side: 150<br>Object side: 215          | Image side: 150<br>Object side: 210 | Image side: 145<br>Object side: 228          | Image side: 143<br>Object side: 178 |

**Table 3.**

**Working distance of uncorrected and corrected microendoscopes based on long GRIN lenses obtained with ray-trace simulations shown in**

**Fig. 1** [↗](#).

To enable fast manufacturing of multiple corrected microendoscopes, we generated replicas of the aspherical lenses using a molding procedure (**Fig. 2C**) as in <sup>28</sup>. We first fabricated negative molds with PDMS (Sylgard 184 Silicone Elastomer Kit, Dow, Midland, MI, US) casted onto the aspherical lens prototype and solidified at room temperature for 48 hours. We then deposited a small drop of optically transparent UV-curable glue (NOA63, Norland Products, Cranbury, NJ, US) to fill the negative mold. Extreme care was used to avoid bubble formation in this experimental step. We placed a round glass coverslip (coverslip diameter: 3 or 5 mm in diameter; coverslip thickness: 100  $\mu$ m) on top of the drop of UV-curable glue and pressured it against the PDMS mold (**Fig. 2C**). NOA63 was solidified with UV light exposure for 10 minutes. After UV curing, the microlens was visually inspected at the stereomicroscope. In case of formation of air bubbles, the microlens was discarded (yield of the molding procedure:  $\sim 90\%$ ,  $N > 30$  molded lenses). The coverslip with the attached corrective lens was sealed to a customized metal or plastic support ring of appropriate diameter (**Fig. 2C**). The support ring, the coverslip and the aspherical lens formed the upper part of the corrected microendoscope, to be subsequently coupled to the proper GRIN rod (**Table 2**) using a custom-built opto-mechanical stage and NOA63 (**Fig. 2C**). The GRIN rod was positioned perpendicularly to the glass coverslip, on the other side of the coverslip compared to the corrective lens, and aligned to the aspherical lens perimeter (**Fig. 2C**) under the guidance of a wide field microscope equipped with a camera. The yield of the assembly procedure for the probes used in this work was 100 % ( $N = 27$  endoscopes). For further details on the assembly of corrected microendoscope see <sup>7</sup>.

### Stylus profilometer measurements on polymeric corrective lenses

To measure the surface of the replicated corrective lenses, we used a stylus profilometer (stylus radius, 5  $\mu$ m; applied “force”, 1 mg; horizontal resolution, 0.056  $\mu$ m; Dektak 150 Profiler, Veeco, Plainview, NY). We measured the height of the surface of multiple polymer corrective lenses for both the microendoscope types. For each corrective lens replica, we measured the profile along three different diameters in order to evaluate potential asymmetries in the lens replication process. The radial profiles obtained for each microendoscope type were aligned and pooled together to calculate the average profile (Supplementary Figure 4).

### Optical characterization of corrected microendoscopes

We performed optical characterization of corrected and uncorrected microendoscopes using 2P laser scanning microscopes equipped with femtosecond tunable laser sources (80 MHz repetition rate, Chameleon Ultra II or Chameleon Discovery, Coherent, Inc., Santa Clara, CA, US) tuned at 920 nm excitation wavelength and coupled to an Ultima IV or an Investigator scan head (Bruker Corporation, Billerica, MA, US). We used a 20x dry objective (EC Epiplan-Neofluar, NA = 0.5, Zeiss, Oberkochen, Germany) and coupled it with the microendoscopes using a custom optomechanical mount similarly to <sup>7</sup>. The mount was equipped with a micrometric translator to adjust the distance between the objective lens and the upper surface of the corrected microendoscope. We acquired series of z-stacks of different calibration samples (subresolved fluorescent films, subresolved fluorescent beads, calibration rulers, and mouse fixed brain slices) moving the whole mount carrying both the objective and the microendoscope, so that their reciprocal distance remained fixed during the acquisition. For all measurements, the distance between the objective and the upper surface of the coverslip was set to 100  $\mu$ m for uncorrected microendoscopes and to 150  $\mu$ m and 180  $\mu$ m for the 6.4 mm-long and the 8.8 mm-long corrected microendoscopes, respectively. This choice was made taking into account the working distances of simulated corrected probes (**Table 3** and **Supplementary Figure 4**). The front end of the GRIN rod was immersed into a drop of distilled water, or phosphate-buffered saline (PBS) for fixed brain slices, which was placed over the calibration sample. Due to the short working distance of the microendoscopes (**Table 3**), we put no coverslip over the samples. For each measurement, experimental replicates were obtained using corrected microendoscopes assembled with distinct optical elements (i.e., different GRIN lenses and different corrective lenses). The uncorrected microendoscopes were assembled either using different optical elements compared to the



corrected ones, or were obtained from the corrected probes after the mechanical removal of the corrective lens. The analysis of z-stacks acquired through calibration samples was performed using ImageJ/Fiji and custom code written in Matlab (R2022b, MathWorks, Natick, MA, US). In the following paragraphs, we defined “nominal” pixel size as the one determined by imaging a calibration ruler (see below) with the microscope objective alone.

## Subresolved fluorescent films

To evaluate the axial extension of the excitation volume for the two types of microendoscopes as a function of the radial distance, we imaged subresolved homogeneous fluorescent layers (thickness: ~ 300 nm) mounted on a microscope slide. We acquired z-stacks at 256 pixels x 256 pixels resolution (nominal pixel size: 1.73 μm/pixel) with 2 μm axial step and 16 frame averaging. To analyze the axial profile of the imaged film, for each z-stack we generated the mean  $x,z$  intensity projection by averaging 180 evenly distributed sections passing through the center of the FOV of the microendoscope. This  $x,z$  fluorescence intensity projection represents the section of the FOV. From this image we then extracted the fluorescence intensity profiles along 100 segments evenly distributed across the FOV and orthogonal to the line that is locally tangent to the film profile. We measured the thickness of the average film profile across the FOV as FWHM of Gaussian curves fitting the extracted fluorescence intensity profiles along the segments. Data were binned along the  $x$  coordinate of the image (bin size: 10 pixels). To set the zero of the  $x$  coordinate, we fitted the longitudinal intensity profile of the section of the film (along the section of the FOV) to a Gaussian curve and we selected the pixel corresponding to the peak. We finally averaged together binned data obtained from multiple subresolved layer acquisitions using different microendoscopes.

## Calibration rulers

To properly calibrate microendoscope images, we used a calibration ruler with ticks spaced every 10 μm along two orthogonal directions (4-dot calibration slide, Cat. No. 1101002300142, Motic, Hong Kong). The ruler was oriented such that the ruler ticks were along the  $x$  and  $y$  directions of the acquired image and the crossing point between the two orthogonal directions of the ruler was positioned at the center of the microendoscope FOV. z-stacks of the ruler were acquired at 512 pixels x 512 pixels resolution (nominal pixel size: 1.06 μm/pixel) with 2 μm axial step and 8 frame averaging. For each z-stack, we generated the maximum intensity projection and identified tick locations as the local maxima of the intensity profiles obtained along two lines drawn along the two ruler directions. We then counted the number of pixels between the identified tick locations and the center of the FOV. Due to the circular symmetry of GRIN lens-based microendoscopes, we averaged together data obtained along the four radii of the ruler. We finally averaged across measurements performed with different microendoscopes. Knowing the average number of pixels in between adjacent ticks in the image and the real distance between adjacent ticks in the sample (10 μm), we could determine, at each tick position, the local pixel size by dividing 10 μm by the mean number of pixels in between two adjacent ticks.

We observed that corrected microendoscopes had uneven magnification across the FOV (distortion), meaning that the local pixel size is not homogeneous all over the FOV. To characterize the uneven magnification independently of the acquisition setting (pixel size and microscope magnification), we computed the local magnification factor. This quantity was used to calibrate the local pixel size as a function of the radial distance. The local magnification factor was defined as the ratio between the nominal pixel size and the local pixel size obtained as described above. The values of the local magnification factor as a function of the radial distance were fitted with a quartic function  $f(x) = ax^4 + bx^2 + c$  for each microendoscope type (uncorrected and corrected) and the fitting function was used to calibrate the pixel size of all acquired images (Supplementary Table 3). For each image acquired using corrected microendoscopes (e.g., Fig. 4), we show the scale bar corresponding to the calibrated local pixel size at the center of the FOV. The radial change of pixel size in the  $x$  and  $y$  directions is visualized by a color-coded bar located close to the

image and representing the local magnification factor normalized to the magnification factor at the center of the FOV. Additionally, considering the cumulative effect of the local change in the pixel size, we performed a calibration of radial distances longer than 10-20  $\mu\text{m}$ , using the same ruler measurements described above. This calibration was obtained by fitting the nominal radial distances at which we identified the ruler ticks as a function of their real distances (given by the ruler spacing) with a quartic function  $g(x) = dx^4 + ex^2 + fx + h$ , for each microendoscope type (**Supplementary Table 3**). The fitting function was then used to calculate the real distance among different structures across the FOV.

## Subresolved fluorescent beads

To quantify the extension of the PSF, we imaged subresolved fluorescent beads (FluoSpheres Carboxylate-Modified Microspheres, diameter: 100 nm, Cat. No. F8803, Thermofisher, Waltham, MA, US) monodispersed in a thin layer of poly-l-lysine (0.1 mg/mL, Cat. No. P2636, Merck, Darmstadt, Germany) on a microscope slide. Multiple z-stacks were acquired at 512 pixels x 512 pixels resolution (nominal pixel size: 0.049  $\mu\text{m}/\text{pixel}$ ) with 0.5  $\mu\text{m}$  axial step and 8 frame averaging. Fluorescent beads were present across the FOV at different nominal radial distances from the optical axis (0-175  $\mu\text{m}$  for corrected probes, 0-75  $\mu\text{m}$  for uncorrected probes). In uncorrected microendoscopes, strong aberrations prevented imaging beads at nominal radial distances > 75  $\mu\text{m}$ . To quantify the axial resolution, for each bead we generated the z-axis intensity profile using the average value of the fluorescence signal within a ROI encompassing the whole bead perimeter after background subtraction and fitted this profile to either one or two Gaussian curves. For single Gaussian curves, we estimated the axial resolution as FWHM of the curve. In the case of fitting with two Gaussian curves, we computed the axial resolution as the sum of the half width at half maximum of the two Gaussian curves and the distance between their peaks. To quantify the lateral resolution, we generated the x,y intensity projections of individual beads and measured the intensity spatial distribution along 8 profiles in 8 different and equally spaced directions centered on the bead. We fitted each intensity profile to a Gaussian curve and we computed the average value of the FWHM of each curve as the lateral resolution. For each microendoscope type, measurements of axial and lateral resolution performed with multiple microendoscopes on multiple beads for each radial distance were pooled together to compute the mean value.

## Mouse fixed brain slices

To prepare brain slices, a deeply anesthetized mouse in which thalamic neurons were expressing jRCaMP7f was transcardially perfused first with PBS and then with 4% paraformaldehyde (PFA). The brain was harvested, fixed in 4% PFA overnight, and kept in sucrose (30 % v/v in PBS). Coronal sections (thickness: 50  $\mu\text{m}$ ) of the brain were cut using a microtome and kept in anti-freezing medium at -20° C. On the day of the experiment, slices were washed three times with PBS for 10 minutes. One slice at a time was gently placed over a microscope slide and kept wet with drops of PBS along the duration of the experiment. On identified FOV containing neurons expressing the fluorescent indicator, we acquired a z-stack through the slice using a corrected or uncorrected microendoscope (1024 pixels x 1024 pixels resolution; nominal pixel size: 0.45  $\mu\text{m}/\text{pixel}$ ; axial step: 5  $\mu\text{m}$ ; number of axial steps: 23-32; frame averaging = 8).

## Generation of synthetic calcium imaging data

### Geometry of simulated neurons

For all simulated neural data (**Fig. 5**, **6**, **Supplementary Figure 5**, **Supplementary Figure 6**), neurons were simulated as spheres with radius randomly sampled from a Gaussian distribution with mean and standard deviation (SD) estimated from the literature to be 10  $\mu\text{m}$  and 3  $\mu\text{m}$ , respectively. A nuclear region, of width randomly sampled from a normal distribution with mean = 5  $\mu\text{m}$  and SD = 1  $\mu\text{m}$  and not expressing the fluorescent indicator, was added at the center

of each simulated neuron. Therefore, only the spherical shell surrounding the nucleus generated the fluorescence signal. Simulated neurons were randomly placed with no overlap in a volume of size  $400 \times 400 \times 170 \mu\text{m}^3$  and  $360 \times 360 \times 200 \mu\text{m}^3$  for the 6.4 mm-long and the 8.8 mm-long microendoscopes, respectively, until neural density reached the value  $(26.8 \pm 7.2) \cdot 10^4$  cells/mm<sup>3</sup><sup>32,29</sup> or the volume was filled up with neurons. The resolution of the spatial volume was  $0.8 \mu\text{m}/\text{pixel}$  in the x and y direction,  $1 \mu\text{m}/\text{pixel}$  in the z direction.

## Simulated neural activity

Neural spiking activity was simulated for 5 minutes in 1 ms time steps. Each neuron was randomly paired with other neurons (probability of two neurons of being paired equal to 0.05) to form groups of neurons with shared activity. The activity of each neuron was then simulated as the sum of an independent Poisson process and as many shared Poisson processes as the number of groups the neuron belonged to. The spike rate of the summed Poisson processes was the same for all neurons (0.3 Hz). These conditions resulted in average pairwise correlations of ground truth calcium activity larger than zero (average correlation  $0.041 \pm 0.001$ ,  $n = 968130$ ). This correlation value was roughly in the order of average pairwise correlations between neurons experimentally recorded with 2P calcium imaging<sup>52,53</sup> and was independent of the radial distance. Calcium activity and fluorescence traces were finally generated using the equations and the parameters described for GCaMP8f in<sup>31</sup>.

## Generation of fluorescence t-series

The simulated FOV had dimensions  $400 \times 400 \mu\text{m}^2$  and  $360 \times 360 \mu\text{m}^2$  for the 6.4 mm-long and the 8.8 mm-long microendoscopes, respectively. The resolution was then adjusted according to the variations of the magnification factor, which were estimated from experimental data in **Figure 3E**, **F**<sup>34</sup>, and **Supplementary Table 3**<sup>34</sup>. This procedure led to a non-uniform resolution across the simulated FOV. To generate the synthetic calcium imaging t-series, we used the experimental measurements of spatial resolution reported in **Fig. 3I, J**<sup>34</sup> for both corrected and uncorrected microendoscopes. For corrected microendoscopes, the 5<sup>th</sup> order polynomial function fitted to the experimental measurements of subresolved fluorescent films was rotated along the z-axis to generate the synthetic imaging focal surface. Moreover, the volume of fluorescence excitation was an ellipsoid resembling the experimentally measured PSF of corrected microendoscopes. For uncorrected microendoscopes, the synthetic imaging surfaces were two spherical shells with curvature radius estimated from the data (curvature radius:  $273 \mu\text{m}$  and  $2000 \mu\text{m}$  for the 6.4 mm-long uncorrected microendoscope and  $992 \mu\text{m}$  and  $2000 \mu\text{m}$  for the 8.8 mm-long uncorrected microendoscope) and the volume of fluorescence excitation was an ellipsoid resembling the experimentally measured PSF of uncorrected microendoscopes.

Simulated excitation volumes were synthetically scanned along the imaging focal surface (or surfaces), such that their axial direction was always orthogonal to the imaging focal surface(s).

Due to the stronger field curvature of the 8.8 mm-long corrected microendoscope (**Figure 1C**<sup>34</sup>) compared to 8.8 mm-long uncorrected microendoscopes, the center of the corrected imaging focal surface resulted at a larger depth in the simulated volume compared to the center of the uncorrected focal surface(s). Therefore, different simulated neurons were sampled in the two cases. We exactly followed methods previously published in<sup>7</sup>. For completeness, the procedure description is repeated here. All voxels within the excitation volumes contributed to the signal of the corresponding pixel in the simulated FOV, as follows:

- If the pixel was at the FOV edge (radial distance  $> 200 \mu\text{m}$  and  $> 180 \mu\text{m}$  for the 6.4 mm-long and the 8.8 mm-long corrected microendoscopes, respectively), its signal was randomly sampled from a normal distribution, with mean and SD estimated from four experimental t-series recorded *in vivo* with the 8.8 mm-long corrected microendoscope and jGCaMP8f

expressed in the mouse piriform cortex (see the paragraph “2P microscopy in the mouse olfactory cortex”). The temporal average intensities of pixels in the edges of the microendoscope FOVs, containing dark noise, were used to fit a Gaussian mixture model (component 1: proportion = 0.35; mean = 137.48; SD = 48.96; component 2: proportion = 0.65; mean = 126.83; SD = 5.02). The SD of the dark noise was assumed to depend on the dark noise mean in a linear way, with parameters estimated from the same experimental data (intercept,  $p_0 = -175.39$ ; linear coefficient,  $p_1 = 1.57$ ). The simulated dark noise was generated with the mean randomly sampled from the Gaussian Mixture Modeling (GMM) distribution and the SD proportional to the mean.

- If the pixel was in the central part of the FOV (radial distance  $\leq 200 \mu\text{m}$  and  $\leq 180 \mu\text{m}$  for the 6.4 mm-long and the 8.8 mm-long microendoscopes, respectively) but no neurons were within the excitation volume, the pixel signal was randomly sampled from a normal distribution with mean and SD estimated from experimental data. The mean intensity of pixels that were neither in the edges nor belonging to ROIs were fitted using a lognormal distribution (mean = 6.54, SD = 0.78) and the best linear fit between the squared root of the mean intensity and the intensity SD was computed ( $p_0 = 29.91$ ,  $p_1 = 7.75$ ). Simulated noise in the FOV was generated as Gaussian noise with mean randomly sampled from the lognormal distribution and SD proportional to the square root of the mean.
- If the pixel was in the central portion of the FOV (radial distance  $\leq 200 \mu\text{m}$  and  $\leq 180 \mu\text{m}$  for the 6.4 mm-long and the 8.8 mm-long microendoscopes, respectively) and at least one neuron was in the excitation volume(s), each voxel in the excitation volume(s) was assigned either Gaussian noise (estimated as in the previous condition) in case no neurons were in that voxel, or the fluorescence intensity of the neuron sampled by that voxel. If a neuron was contained within a voxel, Gaussian noise was also added to its signal. The mean of the added Gaussian noise was zero, while the SD was proportional to the square root of the mean intensity of the voxel, with the coefficients estimated from a linear fit between the square root of the mean intensity and the intensity SD of pixels assigned to ROIs in experimental data ( $p_0 = 322.18$ ,  $p_1 = 0.98$ ). The activity of all the voxels falling within the excitation volume(s) was then averaged to obtain the pixel’s fluorescence intensity. The intensity of each pixel signal was finally modulated as a function of the radial position within the FOV, accordingly to the optical characterization of corrected and uncorrected microendoscopes using the radial intensity obtained by imaging the subresolved fluorescent films (Fig. 3).

In simulations, the imaging rate of the t-series was set to 30 Hz.

## Analysis of simulated calcium t-series

### Detection of cell identities and extraction of activity traces

Analysis of synthetic calcium t-series was performed using CITE-ON <sup>32</sup> and custom code written in Python 3.7 (libraries: NumPy, <https://numpy.org/doc/stable/>; SciPy, <https://docs.scipy.org/doc/scipy/>, Statsmodels, <https://www.statsmodels.org/stable/index.html>).

Specifically, cell bodies were detected and marked with rectangular bounding boxes, using the median intensity projection of the t-series as reference image. Fluorescence intensity traces of single neurons as a function of time,  $F(t)$ , were extracted using a custom modified version of the CITE-ON function *extract*. Within each rectangular bounding box and for each frame, this procedure selected as ROI the group of pixels with a fluorescence intensity value between the 80<sup>th</sup> and the 95<sup>th</sup> percentile of the fluorescence intensity distribution within the considered bounding box.  $F(t)$  was computed frame-by-frame as the difference between the average signal of pixels in each ROI and the background signal. The background was calculated as the average signal of pixels that: i) did not belong to any bounding box; ii) had intensity values higher than the mean

noise value measured in pixels located at the corners of the rectangular image, which do not belong to the circular FOV of the microendoscope; *iii*) had intensity values lower than the maximum value of pixels within the boxes. Activity traces were computed as:

$$\frac{\Delta F}{F_0} = \frac{F(t) - F_0(t)}{F_0(t)},$$

where  $F_0(t)$  is the cell baseline fluorescence at time  $t$  estimated as the 20<sup>th</sup> percentile of the fluorescence distribution of the trace  $F(t)$  in a temporal interval of 10 s centered in  $t$ . The peak SNR of activity traces  $\Delta F/F_0$  was computed as:

$$\text{peak SNR} = \frac{\max(\Delta F/F_0)}{\text{noise}},$$

where  $\max(\Delta F/F_0)$  is the maximum value of the activity trace and noise is the SD of the distribution of values below the 25<sup>th</sup> percentile of the intensity distribution of the entire trace <sup>39</sup>. To determine the position in the FOV of detected cells, we used the coordinates of the centroids of the bounding boxes identified by CITE-ON <sup>32</sup>, expressed in pixels. We converted the centroid coordinates in microns by multiplying the centroid coordinates by the nominal pixel size valid under undistorted conditions and then applied the distance magnification factor to correct for the FOV uneven magnification (see the subparagraph “Calibration rulers”). In **Fig. 5**, the number of detected ROIs and the maximum distance from the center of the FOV at which a cell identity was detected were computed as a function of the peak SNR threshold imposed on activity traces (tested peak SNR threshold values: 0, 5, 10, 11, 12.5, 15, 16.5, 17.5, 20, 22.5, 25, 27.5, 30).

## Pairwise correlation of adjacent neurons

In **Fig. 6**, for each microendoscope type we computed the pairwise correlation of adjacent neurons as Pearson’s correlation between the traces of any pair of nearby detected cells (distance between bounding box centroids  $\leq 25 \mu\text{m}$ ). Pearson’s correlation was computed with the function *numpy.corrcoef*. To evaluate the contamination of pairwise correlation artefactually introduced by the spatial extension of the microendoscope excitation volume in uncorrected or corrected synthetic t-series, we estimated the expected cell pair correlation based on the simulated ground truth calcium activity as described below. We first computed the mean Pearson’s correlation between the activity traces of any possible ground truth source pair for each simulated FOV. For each microendoscope type, we then defined the expected cell pair correlation as the mean Pearson’s correlation of any possible ground truth source pair averaged over different FOV, plus three SD (see **Supplementary Table 5** for parameters). In simulated t-series, any cell pair with pairwise correlation above the expected value could not be attributed to correlations between the ground truth calcium activity of source neurons and was expected to arise only because of the source mixing due to finite optical resolution of microendoscopic probes. Thus, we restricted the analysis to cell pairs with pairwise correlation higher than expected. We pooled together pairwise correlation data from FOV obtained for the same microendoscope type and we computed the percentage of detected adjacent cell pairs that had pairwise correlation higher than the expected cell pair correlation for different peak SNR thresholds. Pairwise correlation of adjacent cell pairs that exceeded the expected cell pair correlation threshold was also analyzed as a function of the radial distance of the cell pair, that was defined as the distance between the center of the FOV and the midpoint of the segment connecting the centroids of the two bounding boxes drawn by CITE-ON <sup>32</sup> (**Supplementary Figure 6A, C**).

## General linear model (GLM) of the contribution of ground truth source to activity traces

For each detected ROI of each simulated FOV, we considered the cellular ground truth sources overlapping in the imaging plane with the rectangular bounding box drawn by CITE-ON for at least 1 pixel. From now on, we define those sources as “ground truth sources contributing to the



extracted activity trace”. To evaluate the precision of microendoscopes in faithfully collecting the activity signals from individual cellular sources and avoiding source mixing, for each detected bounding box we applied a GLM of the ground truth sources contributing to its activity trace:

$$Y_i(t) = \sum_{j=1}^{k_i} a_j X_j(t) + q_i,$$

where:  $Y_i(t)$  is the extracted activity trace for the  $i$ -th bounding box, with  $i = 1, \dots, N$ , with  $N$  number of bounding boxes in the FOV with at least one contributing ground truth source;  $k_i$  is the number of contributing ground truth sources for the  $i$ -th bounding box;  $q_i$  is the  $i$ -th constant term;  $X_j(t)$  is the activity trace of the  $j$ -th contributing ground truth source, with  $j = 1, \dots, k_i$ ; and  $a_j$  is the  $j$ -th linear coefficient computed in the model. The GLM was applied using the methods `statsmodels.api.add_constant` to include the constant term and `statsmodels.genmod.generalized_linear_model.GLM.fit` (link function: *Identity*) to compute the  $k_i$  linear coefficients. We evaluated the purity index of each extracted activity trace  $Y_i(t)$  (i.e. the level of mixing of contributing sources) as:

$$\text{purity index} = \frac{\max_j(a_j^2)}{\sum_{j=1}^{k_i} a_j^2}$$

where  $\max_j$  indicates the maximum value over  $j = 1, \dots, k_i$ . We applied the GLM considering different peak SNR thresholds imposed on detected cell identities ([Supplementary Figure 6B, D](#)).

## Correlation of extracted traces with ground truth source activity

For each detected bounding box with at least one contributing ground truth source and with peak SNR of the extracted trace  $> 10$ , we computed the Pearson's correlation of the extracted trace with the activity trace of each contributing ground truth source. We sorted the sources in descending order of correlation and pooled together correlation values for the first (most correlated) ground truth source ([Fig. 6D, I](#)). Correlation with the first ground truth source was analyzed as a function of the radial distance of the detected cell from the center of the FOV ([Fig. 6E, J](#)).

## Animals

Experimental and surgical protocols were performed in accordance with the Guide of Care and Use of Laboratory Animals (NIH) and were approved by the Institutional Animal Care and Use Committee (IACUC) at Brown University and by the Istituto Italiano di Tecnologia Animal Health Regulatory Committee, by the National Council on Animal Care of the Italian Ministry of Health (authorization # 1134/2015-PR, # 689/2018-PR) and carried out according to the National legislation (D.Lgs. 26/2014) and to the legislation of the European Communities Council Directive (European Directive 2010/63/EU). C57BL/6J mice were crossed to Ai14 mice<sup>54</sup> and male and female heterozygous transgenic offspring of 8-12 weeks of age were used. Mice were maintained with unrestricted access to food and water under a 12-hour light/dark cycle and housed individually after surgery.

## Stereotaxic surgery for viral injections and microendoscope implantation

Viruses (pGP-AAV-syn-jGCaMP8f-WPRE<sup>31</sup> or AAV1-syn-jGCaMP7f-WPRE<sup>51</sup>) were purchased from Addgene and injected using manually controlled pressure injection with a micropipette pulled and a micropipette puller (Sutter Instruments, Novato, CA). Mice were anesthetized with isoflurane with an induction at 3% and then maintained at 1-2% with an oxygen flow rate of  $\sim 1\text{L/min}$  and head-fixed in a stereotactic frame (David Kopf, Tujunga, CA). Mice subcutaneously received buprenorphine slow release (0.05 – 0.1 mg/kg). Eyes were lubricated with an ophthalmic ointment and body temperature was stabilized using a heating pad attached to a temperature

controller. Fur was shaved and the incision site was sterilized with isopropyl alcohol and betadine solution prior to beginning surgical procedures. A 1.0 mm round craniotomy was made using a dental drill centered to the following stereotaxic coordinates: ML, 3.9; AP, 0.3. The virus solution diluted 1/3 in PBS was injected at speed 100 nL/min in 3 different spots (total injected volume: 1  $\mu$ L) using the following coordinates (mm) to target olfactory cortex (ML/AP/DV): 3.85/0.6/-3.8, 3.95/0.3/-3.9, 4.05/0.0/-4.0, all relative to bregma<sup>55</sup>. After 5 minutes, the micropipette was slowly retracted from the brain at a speed of 500  $\mu$ m/min. 30 minutes after viral injections, a corrected microendoscope was implanted above the olfactory cortex. The probe was implanted, centered to the craniotomy, at a speed of 100  $\mu$ m/min until reaching the following coordinate: DV,-3.6. Once placed, the plastic ring of the endoscopic probe was fixed to the skull with Metabond adhesive cement (Parkell Inc., Edgewood, NY). A custom-made aluminum head bar was then attached to the skull using dental cement (Pi-ku-plast HP 36 Precision Pattern Resin, XPdent, Miami, FL). Finally, a protective cap made of Kwik-Sil silicone elastomer (World Precision Instruments, Sarasota, FL) was applied over the lens. Mice were allowed to recover from surgery for at least six weeks prior to imaging experiments.

## Odor delivery

Animals were habituated to the experimenter and the head-fixation set-up for 30 minutes per day for at least two days before performing the imaging experiment. On imaging days, odor stimuli were delivered through a custom built 16-channel olfactometer (Automate Scientific, Berkley, CA) equipped with a mass flow controller that maintained air flow at 1 L/min. The olfactometer solenoids were triggered by a Teensy 3.6 (PJRC). Vacuum was applied inside the 2P microscope isolation box to evacuate residual odor. For all experimental sessions, mice were habituated to the 2P microscope head-fixation set-up for 10 minutes prior to imaging. An odor trial lasted 22 seconds (5 seconds of pre-stimulus baseline, 2 seconds of stimulation, 15 seconds of post-stimulus acquisition) with inter-trial intervals of 10 seconds. Odor stimuli were presented in pseudo-randomized fashion and 8 presentations of each odor were performed in a session. The odor panel consisted of (diluted in mineral oil v/v): acetophenone 1%, amylamine 1%, butyl acetate 1%, ethyl hexanoate 1%, 2-Isobutyl-3-methoxypyrazine 1%,  $\beta$ -Ionone 1%, 2,3-Pentanedione 0.1%, Valeric acid 0.1%. A photoionization detector (miniPID 200B, Aurora Scientific, Canada) was used to confirm reliable odor delivery.

## 2P microscopy in the mouse olfactory cortex

A typical imaging experiment lasted  $\sim$ 1.5 hours per mouse. 2P imaging of the olfactory cortex was performed using an Ultima Investigator laser scanning microscope (Bruker Nano, Inc., Middleton, WI) equipped with a 8 KHz resonance galvanometer and dual GaAsP PMTs (Cat. No. H10770, Hamamatsu, Hamamatsu City, Japan). Approximately 90-150 mW of laser power (at 920 nm, from Chameleon Discovery laser source (Coherent Inc, Santa Clara, CA) was used during imaging, with adjustments in power levels to accommodate varying signals for each mouse. After focusing on the corrective lens surface using epifluorescence microscopy, optical viewing was switched to live view through the 2P laser, and a FOV was located by moving the objective  $\sim$ 100 - 200  $\mu$ m upward. Images were acquired with a Nikon 10X Plan Apochromat Lambda objective (0.45 NA, 4.0 mm WD) using a nominal pixel size of 0.72 – 0.84  $\mu$ m/pixel. jGCaMP8f or jGCaMP7f signals were filtered through an ET-GFP (FITC/CY2) filter set. Acquisition speed was 30 Hz for 512 pixels x 512 pixels images.

## Analysis of *in vivo* 2P calcium imaging data

Analysis of 2P experimental t-series requiring detection of cell identities and trace extraction was performed using CITE-ON<sup>32</sup> and custom code written in Python 3.7, as for the synthetic t-series. Raw 2P t-series were first motion-corrected using the CITE-ON function *motion\_corr*, applied twice in succession when needed. Bounding boxes were detected using the median intensity projection of the motion-corrected t-series as reference. Activity traces, peak SNR and Pearson's correlations

were computed as described for synthetic calcium data. Pairwise correlation of adjacent neurons was explored considering nearby detected cells for which the distance between the bounding boxes centroids was either  $\leq 25 \mu\text{m}$  or  $\leq 30 \mu\text{m}$  (Supplementary Table 6 [↗](#)).

## Statistics

Statistical analysis was performed using Matlab for optical characterization data and using the Python library of statistical functions *scipy.stats* (<https://docs.scipy.org/doc/scipy/reference/stats.html> [↗](#)) for synthetic and experimental calcium imaging data. The number  $n$  of repeated measurements and the number  $m$  of experimental replicates are indicated in figure legends. Normality of data distributions was tested using either Shapiro-Wilk test for small sample sizes ( $n < 50$ ) or D'Agostino-Pearson test ( $n \geq 50$ ). The threshold for statistical significance was set at 0.05 and indicated as follows: \*,  $p < 0.05$ , \*\*,  $p < 0.01$ , \*\*\*,  $p < 0.001$ .  $p$  values higher than 0.05 were considered non significant (n.s.). Data are presented as mean values  $\pm$  s.e.m., unless otherwise stated. Linear regressions were performed using the function *scipy.stats.linregress*. To test the significance of linear fit slope being different from zero, we used the Wald test with  $t$ -distribution in case the fit residuals were normally distributed and the permutation test with the function *stats.permutation\_test* if the fit residuals were not normally distributed. The null distribution of linear coefficients used in the permutation test was built performing multiple linear regressions on a subset of possible permutations of data (permutation type: 'pairings',  $10^4$  permutations for  $n < 500$ ,  $10^5$  permutations for  $n \geq 500$ ). To test the significance of linear fit slopes being different from each other, we used a permutation test in which the observations were randomly assigned to the two samples (corrected or uncorrected). The null distribution was built calculating the difference between the two slopes for a subset of possible permutation of data ( $10^4$  permutations for  $n < 500$ ,  $10^5$  permutations for  $n \geq 500$ ). To test the significance of the difference between two means of non-normally distributed samples, we used a permutation test. In this case, the null distribution was built calculating the difference between the two mean values for a subset of possible permutations of data (permutation type: 'independent',  $10^4$  permutations for  $n < 500$ ,  $10^5$  permutations for  $n \geq 500$ ).

## Data and software availability

The software used in this paper to generate artificial t-series and analyze data is available at: <https://gitlab.iit.it/Chiara.Nardin/longmicroendoscopes> [↗](#). Raw data related to all main figures and all supplementary figures are available from the corresponding author upon request.

## Acknowledgements

We thank Celeste Bortolani, Doriana Debellis, Fabio Moia, and Francesca Succol for technical support and members of the Fellin and Panzeri laboratories for useful discussions and suggestions. This work was supported in part by Horizon 2020 ICT (<https://cordis.europa.eu/project/id/101016787> [↗](#), DEEPER) and Next Generation EU (<https://www.fondazione-fair.it/> [↗](#), FAIR).

## Additional files

**Supplementary Information** [↗](#)

## References

1. Svoboda K., Yasuda R. 2006) **Principles of two-photon excitation microscopy and its applications to neuroscience** *Neuron* **50**:823–839
2. Xu C., Nedergaard M., Fowell D.J., Friedl P., Ji N. 2024) **Multiphoton fluorescence microscopy for in vivo imaging** *Cell* **187**:4458–4487
3. Helmchen F., Denk W. 2005) **Deep tissue two-photon microscopy** *Nat Methods* **2**:932–940
4. Endo K., Kazama H. 2022) **Central organization of a high-dimensional odor space** *Curr. Opin. Neurobiol* **73**
5. Sternson S.M. 2013) **Hypothalamic survival circuits: blueprints for purposive behaviors** *Neuron* **77**:810–824
6. Phelps E.A., Lempert K.M., Sokol-Hessner P. 2014) **Emotion and decision making: multiple modulatory neural circuits** *Annu. Rev. Neurosci* **37**:263–287
7. Antonini A., et al. 2020) **Extended field-of-view ultrathin microendoscopes for high-resolution two-photon imaging with minimal invasiveness** *eLife* **9**
8. Jung J.C., Schnitzer M.J. 2003) **Multiphoton endoscopy** *Opt. Lett* **28**:902–904
9. Jung J.C., Mehta A.D., Aksay E., Stepnoski R., Schnitzer M.J. 2004) **In vivo mammalian brain imaging using one-and two-photon fluorescence microendoscopy** *J. Neurophysiol* **92**:3121–3133
10. Reed W.A., Yan M.F., Schnitzer M.J. 2002) **Gradient-index fiber-optic microprobes for minimally invasive in vivo low-coherence interferometry** *Opt. Lett* **27**:1794–1796
11. Barretto R.P., Messerschmidt B., Schnitzer M.J. 2009) **In vivo fluorescence imaging with high-resolution microlenses** *Nat. Methods* **6**:511–512
12. Moore D.T. 1980) **Gradient-index optics: a review** *Appl. Opt* **19**:1035–1038
13. Jennings J.H., et al. 2019) **Interacting neural ensembles in orbitofrontal cortex for social and feeding behaviour** *Nature* **565**:645–649
14. Jung S., et al. 2022) **A forebrain neural substrate for behavioral thermoregulation** *Neuron* **110**:266–279
15. Roland B., Deneux T., Franks K.M., Bathellier B., Fleischmann A. 2017) **Odor identity coding by distributed ensembles of neurons in the mouse olfactory cortex** *eLife* **6**
16. Pashkovski S.L., et al. 2020) **Structure and flexibility in cortical representations of odour space** *Nature* **583**:253–258
17. Wang P.Y., et al. 2020) **Transient and Persistent Representations of Odor Value in Prefrontal Cortex** *Neuron* **108**:209–224

18. Bortoletto F., Bonoli C., Panizzolo P., Ciubotaru C.D., Mammano F. 2011) **Multiphoton fluorescence microscopy with GRIN objective aberration correction by low order adaptive optics** *PLoS. ONE* **6**:e22321
19. Lee W.M., Yun S.H. 2011) **Adaptive aberration correction of GRIN lenses for confocal endomicroscopy** *Opt. Lett* **36**:4608–4610
20. Wang C., Ji N. 2012) **Pupil-segmentation-based adaptive optical correction of a high-numerical-aperture gradient refractive index lens for two-photon fluorescence endoscopy** *Opt. Lett* **37**:2001–2003
21. Wang C., Ji N. 2013) **Characterization and improvement of three-dimensional imaging performance of GRIN-lens-based two-photon fluorescence endomicroscopes with adaptive optics** *Opt. Express* **21**:27142–27154
22. Piantadosi S.C., et al. 2024) **Holographic stimulation of opposing amygdala ensembles bidirectionally modulates valence-specific behavior via mutual inhibition** *Neuron* **112**:593–610
23. Li Y., et al. 2024) **Geometric transformation adaptive optics (GTAO) for volumetric deep brain imaging through gradient-index lenses** *Nat. Commun* **15**
24. Matz G., Messerschmidt B., Gross H. 2015) **Improved chromatical and field correction of high-NA GRIN-based endoscopic imaging systems for new biophotonics applications** *Proc. SPIE* **9304**
25. Smith W.J. 2008) **Modern Optical Engineering** NY: McGraw-Hill
26. Liberale C., Cojoc G., Candeloro P., Di Fabrizio E.M. 2010) **Micro-Optics Fabrication on top of Optical Fibers Using Two-Photon Lythography** *IEEE Photonic. Tech. L* **22**:474–476
27. Gonzalez-Hernandez D., Varapnickas S., Bertoncini A., Liberale C., Malinauskas M. 2023) **Micro-Optics 3D printed via Multi-Photon Laser Lithography** *Advanced Optical Materials* **11**:2201701–2201720
28. Schaap A., Bellouard Y. 2013) **Molding topologically-complex 3D polymer microstructures from femtosecond laser machined glass** *Opt. Mater. Express* **3**:1428–1437
29. Suzuki N., Bekkers J.M. 2010) **Inhibitory neurons in the anterior piriform cortex of the mouse: classification using molecular markers** *J. Comp Neurol* **518**:1670–1687
30. Suzuki N., Bekkers J.M. 2011) **Two layers of synaptic processing by principal neurons in piriform cortex** *J. Neurosci* **31**:2156–2166
31. Zhang Y., et al. 2023) **Fast and sensitive GCaMP calcium indicators for imaging neural populations** *Nature* **615**:884–891
32. Sita L., et al. 2022) **A deep-learning approach for online cell identification and trace extraction in functional two-photon calcium imaging** *Nat. Commun* **13**
33. Franks K.M., et al. 2011) **Recurrent circuitry dynamically shapes the activation of piriform cortex** *Neuron* **72**:49–56



34. Hagiwara A., Pal S.K., Sato T.F., Wienisch M., Murthy V.N. 2012) **Optophysiological analysis of associational circuits in the olfactory cortex** *Front Neural Circuits* **6**
35. Denk W., Strickler J.H., Webb W.W. 1990) **Two-photon laser scanning fluorescence microscopy** *Science* **248**:73–76
36. Ecker A.S., et al. 2010) **Decorrelated neuronal firing in cortical microcircuits** *Science* **327**:584–587
37. Panzeri S., Moroni M., Sfaai H., Harvey C.D. 2022) **The structures and functions of correlations in neural population codes** *Nat. Rev. Neurosci* **23**:551–567
38. Harris K.D., Quiroga R.Q., Freeman J., Smith S.L. 2016) **Improving data quality in neuronal population recordings** *Nat Neurosci* **19**:1165–1174
39. Brondi M., et al. 2020) **High-Accuracy Detection of Neuronal Ensemble Activity in Two-Photon Functional Microscopy Using Smart Line Scanning** *Cell Rep* **30**:2567–2580
40. Berners-Lee A., Shtrahman E., Grimaud J., Murthy V.N. 2023) **Experience-dependent evolution of odor mixture representations in piriform cortex** *PLoS. Biol* **21**:e3002086
41. Ji N., Milkie D.E., Betzig E. 2010) **Adaptive optics via pupil segmentation for high-resolution imaging in biological tissues** *Nat Meth* **7**:141–147
42. Zong W., et al. 2017) **Fast high-resolution miniature two-photon microscopy for brain imaging in freely behaving mice** *Nat. Methods* **14**:713–719
43. Zong W., et al. 2021) **Miniature two-photon microscopy for enlarged field-of-view, multi-plane and long-term brain imaging** *Nat. Methods* **18**:46–49
44. Zong W., et al. 2022) **Large-scale two-photon calcium imaging in freely moving mice** *Cell* **185**:1240–1256
45. Scott B.B., Brody C.D., Tank D.W. 2013) **Cellular resolution functional imaging in behaving rats using voluntary head restraint** *Neuron* **80**:371–384
46. Kara P., Boyd J.D. 2009) **A micro-architecture for binocular disparity and ocular dominance in visual cortex** *Nature* **458**:627–631
47. Ohki K., Chung S., Ch'ng Y.H., Kara P., Reid R.C. 2005) **Functional imaging with cellular resolution reveals precise micro-architecture in visual cortex** *Nature* **433**:597–603
48. Sadakane O., et al. 2015) **Long-Term Two-Photon Calcium Imaging of Neuronal Populations with Subcellular Resolution in Adult Non-human Primates** *Cell Rep* **13**:1989–1999
49. Ebina T., et al. 2018) **Two-photon imaging of neuronal activity in motor cortex of marmosets during upper-limb movement tasks** *Nat. Commun* **9**
50. Chen T.W., et al. 2013) **Ultrasensitive fluorescent proteins for imaging neuronal activity** *Nature* **499**:295–300
51. Dana H., et al. 2019) **High-performance calcium sensors for imaging activity in neuronal populations and microcompartments** *Nat Methods*

52. Runyan C.A., Piasini E., Panzeri S., Harvey C.D. 2017) **Distinct timescales of population coding across cortex** *Nature* **548**:92–96
53. Valente M., et al. 2021) **Correlations enhance the behavioral readout of neural population activity in association cortex** *Nat. Neurosci* **24**:975–986
54. Madisen L., et al. 2010) **A robust and high-throughput Cre reporting and characterization system for the whole mouse brain** *Nat Neurosci* **13**:133–140
55. Paxinos G., Franklin K.B.J. 2012) **The Mouse Brain in Stereotaxic Coordinates**

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## Editors

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### Andrew King

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### Reviewer #1 (Public review):

Summary:

Sattin, Nardin, and colleagues designed and evaluated corrective microlenses that increase the useable field of view of two long (>6mm) thin (500  $\mu$ m diameter) GRIN lenses used in deep-tissue two-photon imaging. This paper closely follows the thread of earlier work from the same group (esp. Antonini et al, 2020; eLife), filling out the quiver of available extended-field-of-view 2P endoscopes with these longer lenses. The lenses are made by a molding process that appears practical and easy to adopt with conventional two-photon microscopes.

Simulations are used to motivate the benefits of extended field of view, demonstrating that more cells can be recorded, with less mixing of signals in extracted traces, when recorded with higher optical resolution. In vivo tests were performed in piriform cortex, which is difficult to access, especially in chronic preparations.

The design, characterization, and simulations are clear and thorough, but they do not break new ground in optical design or biological application. However, the approach shows much promise, including for applications such as miniaturized GRIN-based microscopes. Readers will largely be interested in this work for practical reasons: to apply the authors' corrected endoscopes to their own research.

Strengths:

The text is clearly written, the ex vivo analysis is thorough and well supported, and the figures are clear. The authors achieved their aims, as evidenced by the images presented, and were able to make measurements from large numbers of cells simultaneously in vivo in a difficult preparation.

The authors did a good job of addressing issues I raised in initial review, including analyses of chromaticity and the axial field of view, descriptions of manufacturing and assembly yield,

explanations in the text of differences between ex vivo and in vivo imaging conditions, and basic analysis of the in vivo recordings relative to odor presentations. They have also shortened the text, reduced repetition, and better motivated their approach in the introduction.

**Weaknesses:**

As discussed in review and nicely simulated by the authors, the large figure error indicated by profilometry (~10  $\mu\text{m}$  in some cases on average) is inconsistent with the optical performance improvements observed, suggesting that those measurements are inaccurate. I see no reason to include these inaccurate measurements.

<https://doi.org/10.7554/eLife.101420.2.sa3>

**Reviewer #2 (Public review):**

In this manuscript, the authors present an approach to correct GRIN lens aberrations, which primarily cause a decrease in signal-to-noise ratio (SNR), particularly in the lateral regions of the field-of-view (FOV), thereby limiting the usable FOV. The authors propose to mitigate these aberrations by designing and fabricating aspherical corrective lenses using ray trace simulations and two-photon lithography, respectively; the corrective lenses are then mounted on the back aperture of the GRIN lens.

This approach was previously demonstrated by the same lab for GRIN lenses shorter than 4.1 mm (Antonini et al., eLife, 2020). In the current work, the authors extend their method to a new class of GRIN lenses with lengths exceeding 6 mm, enabling access to deeper brain regions as most ventral region of the mouse brain. Specifically, they designed and characterized corrective lenses for GRIN lenses measuring 6.4 mm and 8.8 mm in length. Finally, they applied these corrected long micro-endoscopes to perform high-precision calcium signal recordings in the olfactory cortex.

Compared with alternative approaches using adaptive optics, the main strength of this method is that it does not require hardware or software modifications, nor does it limit the system's temporal resolution. The manuscript is well-written, the data are clearly presented, and the experiments convincingly demonstrate the advantages of the corrective lenses.

The implementation of these long corrected micro-endoscopes, demonstrated here for deep imaging in the mouse olfactory bulb, will also enable deep imaging in larger mammals such as rats or marmosets.

**Comments on revisions:**

The authors have clearly addressed all my comments.

<https://doi.org/10.7554/eLife.101420.2.sa2>

**Reviewer #3 (Public review):**

**Summary:**

This work presents the development, characterization and use of new thin microendoscopes (500 $\mu\text{m}$  diameter) whose accessible field of view has been extended by the addition of a corrective optical element glued to the entrance face. Two microendoscopes of different lengths (6.4mm and 8.8mm) have been developed, allowing imaging of neuronal activity in brain regions >4mm deep. An alternative solution to increase the field of view could be to add

an adaptive optics loop to the microscope to correct the aberrations of the GRIN lens. The solution presented in this paper does not require any modification of the optical microscope and can therefore be easily accessible to any neuroscience laboratory performing optical imaging of neuronal activity.

**Strengths:**

(1) The paper is generally clear and well written. The scientific approach is well structured and numerous experiments and simulations are presented to evaluate the performance of corrected microendoscopes. In particular, we can highlight several consistent and convincing pieces of evidence for the improved performance of corrected microendoscopes:

- PSFs measured with corrected microendoscopes 75µm from the centre of the FOV show a significant reduction in optical aberrations compared to PSFs measured with uncorrected microendoscopes.

- Morphological imaging of fixed brain slices shows that optical resolution is maintained over a larger field of view with corrected microendoscopes compared to uncorrected ones, allowing neuronal processes to be revealed even close to the edge of the FOV.

- Using synthetic calcium data, the authors showed that the signals obtained with the corrected microendoscopes have a significantly stronger correlation with the ground truth signals than those obtained with uncorrected microendoscopes.

(2) There is a strong need for high quality microendoscopes to image deep brain regions in vivo. The solution proposed by the authors is simple, efficient and potentially easy to disseminate within the neuroscience community.

**Weaknesses:**

Weaknesses that were present in the first version of the paper were carefully addressed by the authors.

<https://doi.org/10.7554/eLife.101420.2.sa1>

**Author response:**

The following is the authors' response to the original reviews.

***Life Assessment***

*This valuable study builds on previous work by the authors by presenting a potentially key method for correcting optical aberrations in GRIN lens-based micro endoscopes used for imaging deep brain regions. By combining simulations and experiments, the authors show that the obtained field of view is significantly increased with corrected, versus uncorrected microendoscopes. The evidence supporting the claims of the authors is solid, although some aspects of the manuscript should be clarified and missing information provided. Because the approach described in this paper does not require any microscope or software modifications, it can be readily adopted by neuroscientists who wish to image neuronal activity deep in the brain.*

We thank the Referees for their interest in the paper and for the constructive feedback. We have taken the time necessary to address all of their comments, acquiring new data and performing additional analyses. With the inclusion of these new results, we modified four main figures (Figures 1, 6, 7, and 8), added three new Supplementary Figures (Supplementary Figures 1, 2, and 3), and significantly edited the text. Based on the additional work suggested



by the Referees, we believe that we have improved our manuscript, provided missing information, and clarified some aspects of the manuscript, which the Referees pointed our attention to.

### **Public Reviews:**

#### **Reviewer #1 (Public review):**

##### *Summary:*

*Referee's comment: Sattin, Nardin, and colleagues designed and evaluated corrective microlenses that increase the useable field of view of two long (>6mm) thin (500  $\mu$ m diameter) GRIN lenses used in deep-tissue two-photon imaging. This paper closely follows the thread of earlier work from the same group (e.g. Antonini et al, 2020; eLife), filling out the quiver of available extended-field-of-view 2P endoscopes with these longer lenses. The lenses are made by a molding process that appears practical and easy to adopt with conventional two-photon microscopes.*

*Simulations are used to motivate the benefits of extended field of view, demonstrating that more cells can be recorded, with less mixing of signals in extracted traces, when recorded with higher optical resolution. In vivo tests were performed in the piriform cortex, which is difficult to access, especially in chronic preparations.*

*The design, characterization, and simulations are clear and thorough, but not exhaustive (see below), and do not break new ground in optical design or biological application. However, the approach shows much promise, including for applications not mentioned in the present text such as miniaturized GRIN-based microscopes. Readers will largely be interested in this work for practical reasons: to apply the authors' corrected endoscopes.*

##### *Strengths:*

*The text is clearly written, the ex vivo analysis is thorough and well-supported, and the figures are clear. The authors achieved their aims, as evidenced by the images presented, and were able to make measurements from large numbers of cells simultaneously in vivo in a difficult preparation.*

##### *Weaknesses:*

*Referee's comment: (1) The novelty of the present work over previous efforts from the same group is not well explained. What needed to be done differently to correct these longer GRIN lenses?*

We thank the Referee for the positive evaluation of our work. The optical properties of GRIN lenses depend on the geometrical and optical features of the specific GRIN lens type considered, i.e. its diameter, length, numerical aperture, pitch, and radial modulation of the refractive index. Our approach is based on the addition of a corrective optical element at the back end of the GRIN lens to compensate for aberrations that light encounters as it travels through the GRIN lens. The corrective optical element must, therefore, be specifically tailored to the specific GRIN lens type we aim to correct the aberrations of. The novelty of the present article lies in the successful execution of the ray-trace simulations and two-photon lithography fabrication of corrective optical elements necessary to achieve aberration correction in the two novel and long GRIN lens types, i.e. NEM-050-25-15-860-S-1.5p and NEM-050-23-15-860-S-2.0p (GRIN length, 6.4 mm and 8.8 mm, respectively). Our previous work (Antonini et al. eLife 2020) demonstrated aberration correction with GRIN lenses shorter than 4.1 mm. The design and fabrication of a single corrective optical element suitable to enlarge the field-of-view (FOV) in these longer GRIN lenses is not obvious, especially because longer

GRIN lenses are affected by stronger aberrations. To better clarify this point, we revised the Introduction at page 5 (lines 3-10 from bottom) as follows:

“Recently, a novel method based on 3D microprinting of polymer optics was developed to correct for GRIN aberrations by placing specifically designed aspherical corrective lenses at the back end of the GRIN lens 7. This approach is attractive because it is built-in on the GRIN lens and corrected microendoscopes are ready-to-use, requiring no change in the optical set-up. However, previous work demonstrated the feasibility of this method only for GRIN lenses of length < 4.1 mm 7, which are too short to reach the most ventral regions of the mouse brain. The applicability of this technology to longer GRIN lenses, which are affected by stronger optical aberrations 19, remained to be proven.”

*(2) Some strong motivations for the method are not presented. For example, the introduction (page 3) focuses on identifying neurons with different coding properties, but this can be done with electrophysiology (albeit with different strengths and weaknesses). Compared to electrophysiology, optical methods more clearly excel at genetic targeting, subcellular measurements, and molecular specificity; these could be mentioned.*

Thank you for the comment. We added a paragraph in the Introduction (page 3, lines 2-8) according to what suggested by the Reviewer:

“High resolution 2P fluorescence imaging of the awake brain is a fundamental tool to investigate the relationship between the structure and the function of brain circuits 1. Compared to electrophysiological techniques, functional imaging in combination with genetically encoded indicators allows monitoring the activity of genetically targeted cell types, access to subcellular compartments, and tracking the dynamics of many biochemical signals in the brain (2). However, a critical limitation of multiphoton microscopy lies in its limited (< 1 mm) penetration depth in scattering biological media 3”.

*Another example, in comparing microfabricated lenses to other approaches, an unmentioned advantage is miniaturization and potential application to mini-2P microscopes, which use GRIN lenses.*

We added the concept suggested by the Reviewer in the Discussion (page 21, lines 4-7 from bottom). The text now reads:

“Another advantage of long corrected microendoscopes described here over adaptive optics approaches is the possibility to couple corrected microendoscopes with portable 2P microscopes 42-44, allowing high resolution functional imaging of deep brain circuits on an enlarged FOV during naturalistic behavior in freely moving mice”.

*(3) Some potentially useful information is lacking, leaving critical questions for potential adopters:*

*How sensitive is the assembly to decenter between the corrective optic and the GRIN lens?*

Following the Referee’s comment, we conducted new optical simulations to evaluate the decrease in optical performance of the corrected endoscopes as a function of the radial shift of the corrective lens from the optical axis of the GRIN rod (decentering, new Supplementary Figure 3), using light rays passing either off- or on-axis. For off-axis rays, we found that the Strehl ratio remained above 0.8 (Maréchal criterion) for positive translations in the range 6-11.5 microns and 16-50 microns for the 6.4 mm- and the 8.8 mm-long corrected microendoscope, respectively, while the Strehl ratio decreased below 0.8 for negative translations of amplitude ~ 5 microns. Please note that for the most marginal rays, a negative translation produces a mismatch between the corrective microlens and the GRIN lens such

that the light rays no longer pass through the corrective lens. In contrast, rays passing near the optical axis were still focused by the corrected probe with Strehl ratio above 0.8 in a range of radial shifts of -40 – 40 microns for both microendoscope types. Altogether, these novel simulations suggest that decentering between the corrective microlens and the GRIN lens < 5 microns do not majorly affect the optical properties of the corrected endoscopes. These new results are now displayed in Supplementary Figure 3 and described on page 7 (lines 3-5 from bottom).

| *What is the yield of fabrication and of assembly?*

The fabrication yield using molding was ~ 90% (N > 30 molded lenses). The main limitation of this procedure was the formation of air bubbles between the mold negative and the glass coverslip. Molded lenses were visually inspected with a stereomicroscope and, in case of air bubble formation, they were discarded.

The assembly yield, i.e. correct positioning of the GRIN lens with respect to the coverslip, was 100 % (N = 27 endoscopes).

We added this information in the Methods at page 29 (lines 1-12), as follows:

“After UV curing, the microlens was visually inspected at the stereomicroscope. In case of formation of air bubbles, the microlens was discarded (yield of the molding procedure: ~ 90 %, N > 30 molded lenses). The coverslip with the attached corrective lens was sealed to a customized metal or plastic support ring of appropriate diameter (Fig. 2C). The support ring, the coverslip and the aspherical lens formed the upper part of the corrected microendoscope, to be subsequently coupled to the proper GRIN rod (Table 2) using a custom-built opto-mechanical stage and NOA63 (Fig. 2C) 7. The GRIN rod was positioned perpendicularly to the glass coverslip, on the other side of the coverslip compared to the corrective lens, and aligned to the aspherical lens perimeter (Fig. 2C) under the guidance of a wide field microscope equipped with a camera. The yield of the assembly procedure for the probes used in this work was 100 % (N = 27 endoscopes). For further details on the assembly of corrected microendoscope see(7)”.

Supplementary Figure 1: Is this really a good agreement between the design and measured profile? Does the figure error (~10 um in some cases on average) noticeably degrade the image?

As the Reviewer correctly noticed, the discrepancy between the simulated profile and the experimentally measured profile can be up to 5-10 microns at specific radial positions. This discrepancy could be due to issues with: (i) the fabrication of the microlens; (ii) the experimental measurement of the lens profile with the stylus profilometer. To discriminate among these two possibilities, we asked what would be the expected optical properties of the corrected endoscope should the corrective lens have the experimentally measured (not the simulated) profile. To this aim, we performed new optical simulations of the point spread function (PSF) of the corrected probe using, as corrective microlens profile, the average, experimentally measured, profile of a fabricated corrective lens. For both microendoscope types, we first fitted the mean experimentally measured profile of the fabricated lens with the aspherical function reported in equation (1) of the main text:

$$f(r) = cr^2 / \left( 1 + \sqrt{(1 - (1+k)c^2r^2)} \right) + a_1r^2 + a_2r^4 + a_3r^6 + a_4r^8 - z_0 \quad (1)$$

where:

- is the radial distance from the optical axis;

- is equal to  $1/R$ , where  $R$  is the radius of curvature;
- is the conic constant;
- - are asphericity coefficients;
- is the height of the microlens profile on-axis.

The fitting values of the parameters of equation (1) for the two lenses are reported for the Referee's inspection here below (variables describing distances are expressed in mm):

#### Author response table 1.

Fitting values for the parameters of Equation (1) describing the profile of corrective microlens replicas measured with the stylus profilometer. Distances are expressed in mm.

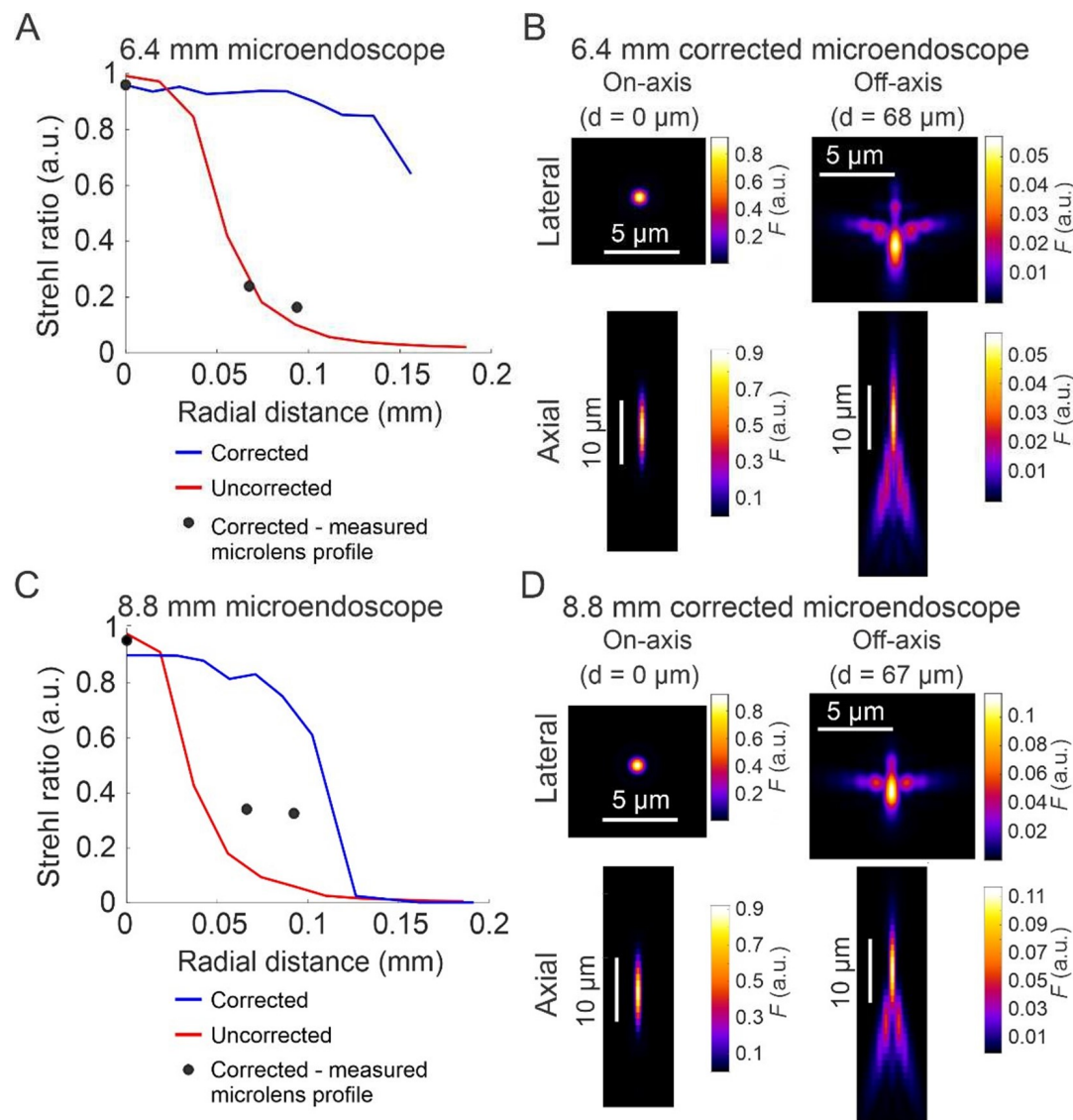
| Fit parameters | 6.4 mm-long microendoscope | 8.8 mm-long microendoscope |
|----------------|----------------------------|----------------------------|
| R              | -2.7229                    | 6.0499                     |
| k              | -1.39                      | 1.6069                     |
|                | -1.0981                    | -1.6162                    |
|                | -29.74                     | -2.4442                    |
|                | 679.7931                   | 443.5147                   |
|                | 999.999                    | 999.9999                   |
|                | 0.012987                   | 0.039158                   |
| Rsquare        | 0.9681                     | 0.9894                     |

We then assumed that the profile of the corrective microlenses were equal to the mean experimentally measured profiles and used the aspherical fitting functions in the optical simulations to compute the performance of corrected microendoscopes. For both microendoscope types, we found that the Strehl ratio was lower than 0.35, well below the theoretical diffractionlimited threshold of 0.8 (Maréchal criterion) at moderate distances from the optical axis (68  $\mu$ m-94  $\mu$ m and 67  $\mu$ m-92  $\mu$ m on the focal plane in the object space, after the front end of the GRIN lens, for the 6.4 mm- and the 8.8 mm-long corrected microendoscope, respectively, Author response image 1A, C), and the PSF was strongly distorted (Author response image 1B, D).

#### Author response image 1.

Simulated optical performance of corrected probes with profiles of corrective microlenses equal to the mean experimentally measured profiles of fabricated corrective lenses. A) The Strehl ratio for the 6.4 mm-long corrected microendoscope with measured microlens profile (black dots) is computed on-axis (distance from the center of the FOV  $d = 0$   $\mu$ m) and at two radial distances off-axis ( $d = 68$   $\mu$ m and  $94$   $\mu$ m on the focal plane in the object space) and compared to the Strehl ratio of the uncorrected (red line) and corrected (blue line) microendoscopes. B) Lateral ( $x,y$ ) and axial ( $x,z$ ) fluorescence intensity ( $F$ ) profiles of

simulated PSFs on-axis (left) and off-axis (right, at the indicated distance  $d$  computed on the focal plane in the object space) for the 6.4 mm-long corrected microendoscope with measured microlens profile. C) Same as in (A) for the 8.8 mm-long corrected microendoscope (off-axis  $d = 67 \mu\text{m}$  and  $92 \mu\text{m}$  on the focal plane in the object space). D) Same as in (B) for the 8.8 mm-long corrected microendoscope.



These simulated findings are in contrast with the experimentally measured optical properties of our corrected endoscopes (Figure 3). In other words, these novel simulated results show that experimentally measured profiles of the corrected lenses are incompatible with the experimental measurements of the optical properties of the corrected endoscopes. Therefore, our experimental recording of the lens profile shown in Supplementary Figure 1 of the first submission (now Supplementary Figure 4) should be used only as a coarse measure of the lens shape and cannot be used to precisely compare simulated lens profiles with measured lens profiles.

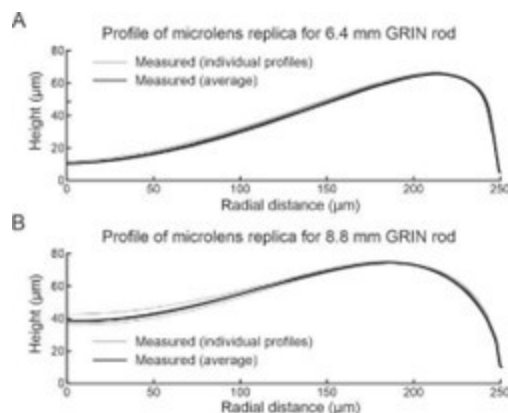
*How do individual radial profiles compare to the presented means?*

We provide below a modified version of Supplementary Figure 4 (Supplementary Figure 1 in the first submission), where individual profiles measured with the stylus profilometer and

the mean profile are displayed for both microendoscope types (Author response image 2). In the manuscript (Supplementary Figure 4), we would suggest to keep showing mean profiles  $\pm$  standard errors of the mean, as we did in the original submission.

## Author response image 2.

Characterization of polymeric corrective lens replicas. A) Stylus profilometer measurements were performed along the radius of the corrective polymer microlens replica for the 6.4 mm-long corrected microendoscope. Individual measured profiles (grey solid lines) obtained from  $n = 3$  profile measurements on  $m = 3$  different corrective lens replicas, plus the mean profile (black solid line) are displayed. B) Same as (A) for the 8.8 mm-long microendoscope.



*What is the practical effect of the strong field curvature? Are the edges of the field, which come very close to the lens surface, a practical limitation?*

A first practical effect of the field curvature is that structures at different  $z$  coordinates are sampled. The observed field curvature of corrected endoscopes may therefore impact imaging in brain regions characterized by strong axially organized anatomy (e.g., the pyramidal layer of the hippocampus), but would not significantly affect imaging in regions with homogeneous cell density within the axial extension of the field curvature ( $< 170 \mu\text{m}$ , see more details below). A second consequence of the field curvature, as the Referee correctly points out, is that cell at the border of the FOV are closer to the front end of the GRIN lens. In measurements of subresolved fluorescent layers (Figure 3A-D), we observed that the field curvature extends in the axial direction to  $\sim 110 \mu\text{m}$  and  $\sim 170 \mu\text{m}$  for the 6.4 mm- and the 8.8 mm-long microendoscopes, respectively. Considered that the nominal working distances on the object side of the 6.4 mm- and the 8.8 mm-long microendoscopes were, respectively,  $210 \mu\text{m}$  and  $178 \mu\text{m}$  (Table 3), structures positioned at the very edge of the FOV were  $\sim 100 \mu\text{m}$  and  $\sim 8 \mu\text{m}$  away from the GRIN front end for the 6.4 mm-long and for the 8.8 mm-long probe, respectively. Previous studies have shown that brain tissue within  $50\text{-}100 \mu\text{m}$  from the GRIN front end may show signs of tissue reaction to the implant (Curreli et al. PLOS Biology 2022, Attardo et al. Nature 2015). Therefore, structures at the very edge of the FOV of the 8.8 mm-long endoscopes, but not those at the edge of the 6.4 mm-long endoscopes, may be within the volume showing tissue reaction. We added a paragraph in the text to discuss these points (page 18 lines 10-14).

*The lenses appear to be corrected for monochromatic light; high-performance microscopes are generally achromatic. Is the bandwidth of two-photon excitation sufficient to warrant optimization over multiple wavelengths?*



Thanks for this comment. All optical simulations described in the first submission were performed at a fixed wavelength ( $\lambda = 920$  nm). Following the Referee's request, we explored the effect of changing wavelength on the Strehl ratio using new optical simulations. We found that the Strehl ratio remains  $> 0.8$  at least within  $\pm 10$  nm from  $\lambda = 920$  nm (new Supplementary Figure 1A-D, left panels), which covers the limited bandwidth of our femtosecond laser. Moreover, these simulations demonstrate that, on a much wider wavelength range (800 - 1040 nm), high Strehl ratio is obtained, but at different  $z$  planes (new Supplementary Figure 1A-D, right panels). This means that the corrective lens is working as expected also for wavelengths which are different from 920 nm, with different wavelengths having the most enlarged FOV located at different working distances. These new results are now described on page 7 (lines 8-10).

*GRIN lenses are often used to access a 3D volume by scanning in  $z$  (including in this study). How does the corrective lens affect imaging performance over the 3D field of view?*

The optical simulations we did to design the corrective lenses were performed maximizing aberration correction only in the focal plane of the endoscope. Following the Referee's comment, we explored the effect of aberration correction outside the focal plane using new optical simulations. In corrected endoscopes, we found that for off-axis rays (radial distance from the optical axis  $> 40$   $\mu\text{m}$ ) the Strehl ratio was  $> 0.8$  (Maréchal criterion) in a larger volume compared to uncorrected endoscopes (new Supplementary Figure 2), demonstrating that the aberration correction method developed in this study does extend beyond the focal plane for short distances. For example, at a radial distance of  $\sim 90$   $\mu\text{m}$  from the optical axis, the axial range in which the Strehl ratio was  $> 0.8$  in corrected endoscopes was 28  $\mu\text{m}$  and 19  $\mu\text{m}$  for the 6.4 mm- and the 8.8 mm-long microendoscope, respectively. These new results are now described on page 7 (10-19).

*(4) The in vivo images (Figure 7D) have a less impressive resolution and field than the ex vivo images (Figure 4B), and the reason for this is not clear. Given the difference in performance, how does this compare to an uncorrected endoscope in the same preparation? Is the reduced performance related to uncorrected motion, field curvature, working distance, etc?*

In comparing images in Figure 4B with images shown in Figure 7D, the following points should be considered:

- (1) Figure 4B is a maximum fluorescence intensity projection of multiple axial planes of a  $z$ -stack acquired through a thin brain slice (slice thickness: 50  $\mu\text{m}$ ) using 8 frame averages for each plane. In contrast, images in Figure 7D are median projection of a  $t$ -series acquired on a single plane in the awake mouse at 30 Hz resonant scanning imaging (8 min, 14,400 frames).
- (2) Images of the fixed brain slice in Figure 4B were acquired at 1024 pixels x 1024 pixels resolution, nominal pixel size 0.45  $\mu\text{m}/\text{pixel}$ , and with objective NA = 0.50, whereas *in vivo* images in Figure 7D were acquired at 512 pixels x 512 pixels resolution, nominal pixel size 0.72 - 0.84  $\mu\text{m}/\text{pixel}$ , and with objective NA = 0.45.
- (3) In the *in vivo* preparation (Figure 7D), excitation and emission light travel through  $> 180$   $\mu\text{m}$  of scattering and absorbing brain tissue, reducing spatial resolution and the SNR of the collected fluorescence signal.
- (4) By shifting the sample in the  $x, y$  plane, in Figure 4B we could chose a FOV containing homogenously stained cells.  $x, y$  shifting and selecting across multiple FOVs was not possible *in vivo*, as the GRIN lens was cemented on the animal skull.

(5) Images in Figure 7D were motion corrected, but we cannot exclude that part of the decrease in resolution observed in Figure 7D when compared to images in Figure 4B are due to incomplete correction of motion artifacts.

For all the reasons listed above, we believe that it is expected to see smaller resolution and contrast in images recorded in vivo (Figure 7D) compared to images acquired in fixed tissue (Figure 4B).

Regarding the question of how do images from an uncorrected and a corrected endoscopes compared in vivo, we think that this comparison is better performed in fixed tissue (Figure 4) or in simulated calcium data (Figure 5-6), rather than in vivo recordings (Figure 7). In fact, in the brain of living mice motion artifacts, changes in fluorophore expression level, variation in the optical properties of the brain (e.g., the presence of a blood vessel over the FOV) may make the comparison of images acquired with uncorrected and corrected microendoscopes difficult, requiring a large number of animals to cancel out the contributions of these factors. Comparing optical properties in fixed tissue is, in contrast, devoid of these confounding factors. Moreover, the major advantage of quantifying how the optical properties of uncorrected and corrected endoscopes impact on the ability to extract information about neuronal activity in simulated calcium data is that, under simulated conditions, we can count on a known ground truth as reference (e.g., how many neurons are in the FOV, where they are, and which is their electrical activity). This is clearly not possible in the in vivo recordings.

*Regarding Figure 7, there is no analysis of the biological significance of the calcium signals or even a description of where olfactory stimuli were presented.*

We appreciate the Reviewer pointing out the lack of detailed analysis regarding the biological significance of the calcium signals and the presentation of olfactory stimuli in Figure 7. Our initial focus was on demonstrating the effectiveness of the optimized GRIN lenses for imaging deep brain areas like the piriform cortex, with an emphasis on the improved signal-to-noise ratio (SNR) these lenses provide. However, we agree that including more context about the experimental conditions would enhance the manuscript. To address this point, we added a new panel (Figure 7F) showing calcium transients aligned with the onset of olfactory stimulus presentations, which are now indicated by shaded light blue areas. Additionally, we have specified the timing of each stimulus presented in Figure 7E. This revision allows readers to better understand the relationship between the calcium signals and the olfactory stimuli.

*The timescale of jGCaMP8f signals in Figure 7E is uncharacteristically slow for this indicator (compared to Zhang et al 2023 (Nature)), though perhaps this is related to the physiology of these cells or the stimuli.*

Regarding the timescale of the calcium signals observed in Figure 7E, we apologize for the confusion caused by a mislabeling we inserted in the original manuscript. The experiments presented in Figure 7 were conducted using jGCaMP7f, not jGCaMP8f as previously stated (both indicators were used in this study but in separate experiments). We have corrected this error in the Results section (caption of Figure 7D, E). It is important to note that jGCaMP7f has a longer half-decay time compared to jGCaMP8f, which could in part account for the slower decay kinetics observed in our data. Furthermore, the prolonged calcium signals can be attributed to the physiological properties of neurons in the piriform cortex. Upon olfactory stimulation, these neurons often fire multiple action potentials, resulting in extended calcium transients that can last several seconds. This sustained activity has been documented in previous studies, such as Roland et al. (eLife 2017, Figure 1C therein) in anesthetized animals and Wang et al. (Neuron 2020, Figure 1E therein) in awake animals, which report similar durations for calcium signals.

*(5) The claim of unprecedented spatial resolution across the FOV (page 18) is hard to evaluate and is not supported by references to quantitative comparisons. The promises of the method for future studies (pages 18-19) could also be better supported by analysis or experiment, but these are minor and to me, do not detract from the appeal of the work.*

GRIN lens-based imaging of piriform cortex in the awake mouse had already been done in Wang et al., Neuron 2020. The GRIN lens used in that work was NEM-050-50-00920-S-1.5p (GRINTECH, length: 6.4 mm; diameter: 0.5 mm), similar to the one that we used to design the 6.4 mm-long corrected microendoscope. Here we used a microendoscope specifically design to correct off-axis aberrations and enlarge the FOV, in order to maximize the number of neurons recorded with the highest possible spatial resolution, while keeping the tissue invasiveness to the minimum. Following the Referee's comments, we revised the sentence at page 19 (lines 68 from bottom) as follows:

"We used long corrected microendoscopes to measure population dynamics in the olfactory cortex of awake head-restrained mice with unprecedented combination of high spatial resolution across the FOV and minimal invasiveness(17)".

*(6) The text is lengthy and the material is repeated, especially between the introduction and conclusion. Consolidating introductory material to the introduction would avoid diluting interesting points in the discussion.*

We thank the Reviewer for this comment. As suggested, we edited the Introduction and shortened the Discussion.

#### **Reviewer #2 (Public review):**

*In this manuscript, the authors present an approach to correct GRIN lens aberrations, which primarily cause a decrease in signal-to-noise ratio (SNR), particularly in the lateral regions of the field-of-view (FOV), thereby limiting the usable FOV. The authors propose to mitigate these aberrations by designing and fabricating aspherical corrective lenses using ray trace simulations and two-photon lithography, respectively; the corrective lenses are then mounted on the back aperture of the GRIN lens.*

*This approach was previously demonstrated by the same lab for GRIN lenses shorter than 4.1 mm (Antonini et al., eLife, 2020). In the current work, the authors extend their method to a new class of GRIN lenses with lengths exceeding 6 mm, enabling access to deeper brain regions as most ventral regions of the mouse brain. Specifically, they designed and characterized corrective lenses for GRIN lenses measuring 6.4 mm and 8.8 mm in length. Finally, they applied these corrected long micro-endoscopes to perform high-precision calcium signal recordings in the olfactory cortex.*

*Compared with alternative approaches using adaptive optics, the main strength of this method is that it does not require hardware or software modifications, nor does it limit the system's temporal resolution. The manuscript is well-written, the data are clearly presented, and the experiments convincingly demonstrate the advantages of the corrective lenses.*

*The implementation of these long corrected micro-endoscopes, demonstrated here for deep imaging in the mouse olfactory bulb, will also enable deep imaging in larger mammals such as rats or marmosets.*

We thank the Referee for the positive comments on our study. We address the points indicated by the Referee in the "Recommendation to the authors" section below.

### Reviewer #3 (Public review):

#### Summary:

*This work presents the development, characterization, and use of new thin microendoscopes (500 $\mu$ m diameter) whose accessible field of view has been extended by the addition of a corrective optical element glued to the entrance face. Two microendoscopes of different lengths (6.4mm and 8.8mm) have been developed, allowing imaging of neuronal activity in brain regions >4mm deep. An alternative solution to increase the field of view could be to add an adaptive optics loop to the microscope to correct the aberrations of the GRIN lens. The solution presented in this paper does not require any modification of the optical microscope and can therefore be easily accessible to any neuroscience laboratory performing optical imaging of neuronal activity.*

#### Strengths:

*(1) The paper is generally clear and well-written. The scientific approach is well structured and numerous experiments and simulations are presented to evaluate the performance of corrected microendoscopes. In particular, we can highlight several consistent and convincing pieces of evidence for the improved performance of corrected microendoscopes:*

*a) PSFs measured with corrected micro endoscopes 75 $\mu$ m from the centre of the FOV show a significant reduction in optical aberrations compared to PSFs measured with uncorrected micro endoscopes.*

*b) Morphological imaging of fixed brain slices shows that optical resolution is maintained over a larger field of view with corrected micro endoscopes compared to uncorrected ones, allowing neuronal processes to be revealed even close to the edge of the FOV.*

*c) Using synthetic calcium data, the authors showed that the signals obtained with the corrected microendoscopes have a significantly stronger correlation with the ground truth signals than those obtained with uncorrected microendoscopes.*

*(2) There is a strong need for high-quality micro endoscopes to image deep brain regions in vivo. The solution proposed by the authors is simple, efficient, and potentially easy to disseminate within the neuroscience community.*

#### Weaknesses:

*(1) Many points need to be clarified/discussed. Here are a few examples:*

*a) It is written in the methods: "The uncorrected microendoscopes were assembled either using different optical elements compared to the corrected ones or were obtained from the corrected*

*probes after the mechanical removal of the corrective lens."*

*This is not very clear: the uncorrected microendoscopes are not simply the unmodified GRIN lenses?*

We apologize for not been clear enough on this point. Uncorrected microendoscopes are not simply unmodified GRIN lenses, rather they are GRIN lenses attached to a round glass coverslip (thickness: 100  $\mu$ m). The glass coverslip was included in ray-trace optical simulations of the uncorrected system and this is the reason why commercial GRIN lenses and corresponding uncorrected microendoscopes have different working distances, as

reported in Tables 2-3. To make the text clearer, we added the following sentence at page 27 (last 4 lines):

“To evaluate the impact of corrective microlenses on the optical performance of GRIN-based microendoscopes, we also simulated uncorrected microendoscopes composed of the same optical elements of corrected probes (glass coverslip and GRIN rod), but in the absence of the corrective microlens”.

*b) In the results of the simulation of neuronal activity (Figure 5A, for example), the neurons in the center of the FOV have a very large diameter (of about 30µm). This should be discussed.*

Thanks for this comment. In synthetic calcium imaging t-series, cell radii were randomly sampled from a Gaussian distribution with mean = 10 µm and standard deviation (SD) = 3 µm. Both values were estimated from the literature (ref. no. 28: Suzuki & Bekkers, *Journal of Neuroscience*, 2011) as described in the Methods (page 35). In the image shown in Figure 5A, neurons near to the center of the FOV have radius of ~ 20 µm corresponding to the right tail of the distribution (mean + 3SD = 19 µm). It is also important to note that, for corrected microendoscopes, neurons in the central portion of the FOV appear larger than cells located near the edges of the FOV, because the magnification depends on the distance from the optical axis (see Figure 3E, F) and near the center the magnification is > 1 for both microendoscope types.

*Also, why is the optical resolution so low on these images?*

Images shown in Figure 5 are median fluorescence intensity projections of 5 minute-long simulated t-series. Simulated calcium data were generated with pixel size 0.8 µm/pixel and frame rate 30 Hz, similarly to *in vivo* recordings. In the simulations, pixels not belonging to any cell soma were assigned a value of background fluorescence randomly sampled from a normal distribution with mean and standard deviation estimated from experimental data, as described in the Methods section (page 37). To simulate activity, the mean spiking rate of neurons was set to 0.3 Hz, thus in a large fraction of frames neurons do not show calcium transients. Therefore, the median fluorescence intensity value of somata will be close to their baseline fluorescence value ( $F_0$ ). Since in simulations  $F_0$  values (~ 45-80 a.u.) were not much higher than the background fluorescence level (~ 45 a.u.), this may generate the appearance of low contrast image in Figure 5A. Finally, we suspect that PDF rendering also contributed to degrade the quality of those images. We will now submit high resolution images alongside the PDF file.

*c) It seems that we can't see the same neurons on the left and right panels of Figure 5D. This should be discussed.*

The Referee is correct. When we intersected the simulated 3D volume of ground truth neurons with the focal surface of microendoscopes, the center of the FOV for the 8.8 mm-long corrected microendoscope was located at a larger depth than the FOV of the 8.8 mm uncorrected microendoscope. This effect was due to the larger field curvature of corrected 8.8 mm-long endoscopes compared to 8.8 mm-long uncorrected endoscopes. This is the reason why different neurons were displayed for uncorrected and corrected endoscopes in Figure 5D. We added this explanation in the text at page 37 (lines 1-4). The text reads:

“Due to the stronger field curvature of the 8.8 mm-long corrected microendoscope (Figure 1C) compared to 8.8 mm-long uncorrected microendoscopes, the center of the corrected imaging focal surface resulted at a larger depth in the simulated volume compared to the center of the uncorrected focal surface(s). Therefore, different simulated neurons were sampled in the two cases”.

d) It is not very clear to me why in Figure 6A, F the fraction of adjacent cell pairs that are more correlated than expected increases as a function of the threshold on peak SNR. The authors showed in Supplementary Figure 3B that the mean purity index increases as a function of the threshold on peak SNR for all micro endoscopes. Therefore, I would have expected the correlation between adjacent cells to decrease as a function of the threshold on peak SNR. Similarly, the mean purity index for the corrected short microendoscope is close to 1 for high thresholds on peak SNR: therefore, I would have expected the fraction of adjacent cell pairs that are more correlated than expected to be close to 0 under these conditions. It would be interesting to clarify these points.

Thanks for raising this point. We defined the fraction of adjacent cell pairs more correlated than expected as the number of adjacent cell pairs more correlated than expected divided by the number of adjacent cell pairs. The reason why this fraction raises as a function of the SNR threshold is shown in Supplementary Figure 2 in the first submission (now Supplementary Figure 5). There, we separately plotted the number of adjacent cell pairs more correlated than expected (numerator) and the number of adjacent cell pairs (denominator) as a function of the SNR threshold. For both microendoscope types, we observed that the denominator more rapidly decreased with peak SNR threshold than the numerator. Therefore, the fraction of adjacent cell pairs more correlated than expected increases with the peak SNR threshold.

To understand why the denominator decreases with SNR threshold, it should be considered that, due to the deterioration of spatial resolution and attenuation of fluorescent signal collection as a function of the radial distance from the optical axis (see for example fluorescent film profiles in Figure 3A, C), increasing the threshold on the peak SNR of extracted calcium traces implies limiting cell detection to those cells located within smaller distance from the center of the FOV. This information is shown in Figure 5C, F.

In the manuscript text, this point is discussed at page 12 (lines 1-3 from bottom) and page 13 (lines 1-4):

“The fraction of pairs of adjacent cells (out of the total number of adjacent pairs) whose activity correlated significantly more than expected increased as a function of the SNR threshold for corrected and uncorrected microendoscopes of both lengths (Fig. 6A, F). This effect was due to a larger decrease of the total number of pairs of adjacent cells as a function of the SNR threshold compared to the decrease in the number of pairs of adjacent cells whose activity was more correlated than expected (Supplementary Figure 5)”.

e) Figures 6C, H: I think it would be fairer to compare the uncorrected and corrected endomicroscopes using the same effective FOV.

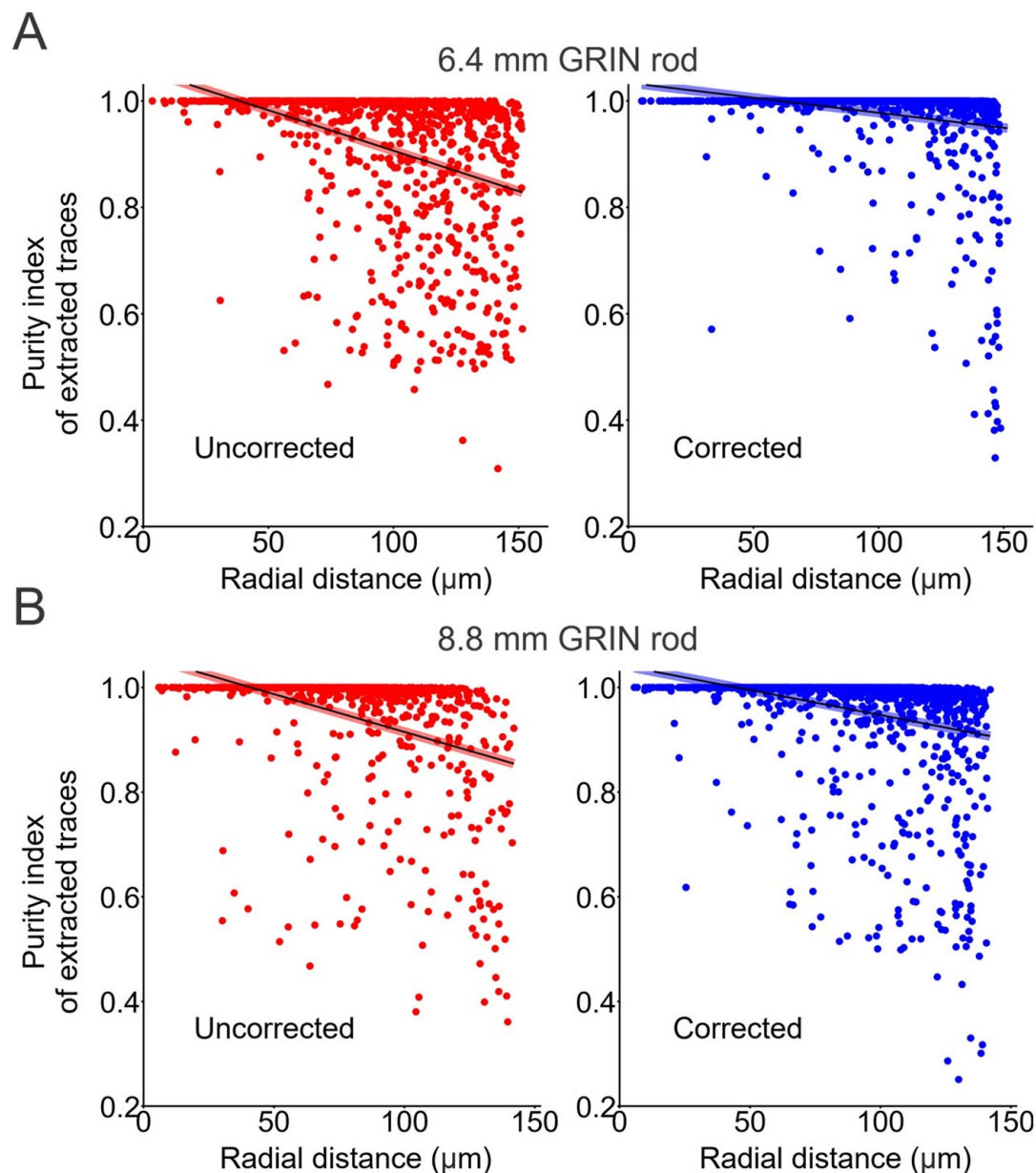
To address the Reviewer’s concern, we repeated the linear regression of purity index as a function of the radial distance using the same range of radial distances for the uncorrected and corrected case of both microendoscope types. Below, we provide an updated version of Figure 6C, H for the referee’s perusal. Please note that the maximum value displayed on the x-axis of both graphs is now corresponding to the minimum value between the two maximum radial distance values obtained in the uncorrected and corrected case (maximum radial distance displayed: 151.6  $\mu\text{m}$  and 142.1  $\mu\text{m}$  for the 6.4 mm- and the 8.8 mm-long GRIN rod, respectively). Using the same effective FOV, we found that the purity index drops significantly more rapidly with the radial distance for uncorrected microendoscopes compared to the corrected ones, similarly to what observed in the original version of Figure 6. The values of the linear regression parameters and statistical significance of the difference between the slopes in the uncorrected and corrected cases are stated in the Author response image 3



caption below for both microendoscope types. In the manuscript, we would suggest to keep showing data corresponding to all detected cells, as we did in the original submission.

### Author response image 3.

Linear regression of purity index as a function of the radial distance. A) Purity index of extracted traces with peak SNR > 10 was estimated using a GLM of ground truth source contributions and plotted as a function of the radial distance of cell identities from the center of the FOV for  $n = 13$  simulated experiments with the 6.4 mm-long uncorrected (red) and corrected (blue) microendoscope. Black lines represent the linear regression of data  $\pm$  95% confidence intervals (shaded colored areas). Maximum value of radial distance displayed: 151.6  $\mu\text{m}$ . Slopes  $\pm$  standard error (s.e.): uncorrected,  $(-0.0015 \pm 0.0002) \mu\text{m}^{-1}$ ; corrected,  $(-0.0006 \pm 0.0001) \mu\text{m}^{-1}$ . Uncorrected,  $n = 991$ ; corrected,  $n = 1156$ . Statistical comparison of slopes,  $p < 10^{-10}$ , permutation test. B) Same as (A) for  $n = 15$  simulated experiments with the 8.8 mm-long uncorrected and corrected microendoscope. Maximum value of radial distance displayed: 142.1  $\mu\text{m}$ . Slopes  $\pm$  s.e.: uncorrected,  $(-0.0014 \pm 0.0003) \mu\text{m}^{-1}$ ; corrected,  $(-0.0010 \pm 0.0002) \mu\text{m}^{-1}$ . Uncorrected,  $n = 718$ ; corrected,  $n = 1328$ . Statistical comparison of slopes,  $p = 0.0082$ , permutation test.



f) Figure 7E: Many calcium transients have a strange shape, with a very fast decay following a plateau or a slower decay. Is this the result of motion artefacts or analysis artefacts?

Thank you for raising this point about the unusual shapes of the calcium transients in Figure 7E. The observed rapid decay following a plateau or a slower decay is indeed a result of how the data were presented in the original submission. Our experimental protocol consisted of 22 s-long trials with an inter-trial interval of 10 s (see Methods section, page 44). In the original figure, data from multiple trials were concatenated, which led to artefactual time courses and apparent discontinuities in the calcium signals. To resolve this issue, we revised Figure 7E to accurately represent individual concatenated trials. We also added a new panel (please see new Figure 7F) showing examples of single cell calcium responses in individual trials without concatenation, with annotations indicating the timing and identity of presented olfactory stimuli.

*Also, the duration of many calcium transients seems to be long (several seconds) for GCaMP8f. These points should be discussed.*

Author response: regarding the timescale of the calcium signals observed in Figure 7E, we apologize for the confusion caused by a mislabeling we inserted in the manuscript. The experiments presented in Figure 7 were conducted using jGCaMP7f, not jGCaMP8f as previously stated (both indicators were used in this study, but in separate experiments). We have corrected this error in the Results section (caption of Figure 7D, E). It is important to note that jGCaMP7f has a longer half-decay time compared to jGCaMP8f, which could in part account for the slower decay kinetics observed in our data. Furthermore, the prolonged calcium signals can be attributed to the physiological properties of neurons in the piriform cortex. Upon olfactory stimulation, these neurons often fire multiple action potentials, resulting in extended calcium transients that can last several seconds. This sustained activity has been documented in previous studies, such as Roland et al. (eLife 2017, Figure 1C therein) in anesthetized animals and Wang et al. (Neuron 2020, Figure 1E therein) in awake animals, which report similar durations for calcium signals. We cite these references in the text. We believe that these revisions and clarifications address the Reviewer's concern and enhance the overall clarity of our manuscript.

*g) The authors do not mention the influence of the neuropil on their data. Did they subtract the neuropil's contribution to the signals from the somata? It is known from the literature that the presence of the neuropil creates artificial correlations between neurons, which decrease with the distance between the neurons (Grødem, S., Nymoen, I., Vatne, G.H. et al. An updated suite of viral vectors for in vivo calcium imaging using intracerebral and retro-orbital injections in male mice. Nat Commun 14, 608 (2023). <https://doi.org/10.1038/s41467-023-363243>; Keemink SW, Lowe SC, Pakan JMP, Dylida E, van Rossum MCW, Rochefort NL. FISSA: A neuropil decontamination toolbox for calcium imaging signals. Sci Rep. 2018 Feb 22;8(1):3493.*

*doi: 10.1038/s41598-018-21640-2. PMID: 29472547; PMCID: PMC5823956*

*This point should be addressed.*

We apologize for not been clear enough in our previous version of the manuscript. The neuropil was subtracted from calcium traces both in simulated and experimental data. Please note that instead of using the term “neuropil”, we used the word “background”. We decided to use the more general term “background” because it also applies to the case of synthetic calcium tseries, where neurons were modeled as spheres devoid of processes. The background subtraction is described in the Methods on page 39:

“ $F(t)$  was computed frame-by-frame as the difference between the average signal of pixels in each ROI and the background signal. The background was calculated as the average signal of pixels that: i) did not belong to any bounding box; ii) had intensity values higher than the mean noise value measured in pixels located at the corners of the rectangular image, which do not belong to the circular FOV of the microendoscope; iii) had intensity values lower than the maximum value of pixels within the boxes”.

*h) Also, what are the expected correlations between neurons in the piriform cortex? Are there measurements in the literature with which the authors could compare their data?*

We appreciate the reviewer's interest in the correlations between neurons in the piriform cortex. The overall low correlations between piriform neurons we observed (Figure 8) are consistent with a published study describing ‘near-zero noise correlations during odor inhalation’ in the anterior piriform cortex of rats, based on extracellular recordings (Miura et

al., Neuron 2013). However, to the best of our knowledge, measurements directly comparable to ours have not been described in the literature. Recent analyses of the correlations between piriform neurons were restricted to odor exposure windows, with the goal to quantify odor-specific activation patterns (e.g. Roland et al., eLife 2017; Bolding et al., eLife 2017, Pashkovski et al., Nature 2020; Wang et al., Neuron 2020). Here, we used correlation analyses to characterize the technical advancement of the optimized GRIN lens-based endoscopes. We showed that correlations of pairs of adjacent neurons were independent from radial distance (Figure 8B), highlighting homogeneous spatial resolution in the field of view.

*(2) The way the data is presented doesn't always make it easy to compare the performance of corrected and uncorrected lenses. Here are two examples:*

*a) In Figures 4 to 6, it would be easier to compare the FOVs of corrected and uncorrected lenses if the scale bars (at the centre of the FOV) were identical. In this way, the neurons at the centre of the FOV would appear the same size in the two images, and the distances between the neurons at the centre of the FOV would appear similar. Here, the scale bar is significantly larger for the corrected lenses, which may give the illusion of a larger effective FOV.*

We appreciate the Referee's comment. Below, we explain why we believe that the way we currently present imaging data in the manuscript is preferable:

(1) current figures show images of the acquired FOV as they are recorded from the microscope (raw data), without rescaling. In this way, we exactly show what potential users will obtain when using a corrected microendoscope.

(2) In the current version of the figures, the fact that the pixel size is not homogeneous across the FOV, nor equal between uncorrected and corrected microendoscopes, is initially shown in Figure 3E, F and then explicitly stated throughout the manuscript when images acquired with a corrected microendoscope are shown.

(3) Rescaling images acquired with the corrected endoscopes gives the impression that the acquisition parameters were different between acquisitions with the corrected and uncorrected microendoscopes, which was not the case.

Importantly, the larger FOV of the corrected microendoscope, which is one of the important technological achievements presented in this study, can be appreciated in the images regardless of the presentation format.

*b) In Figures 3A-D it would be more informative to plot the distances in microns rather than pixels. This would also allow a better comparison of the micro endoscopes (as the pixel sizes seem to be different for the corrected and uncorrected micro endoscopes).*

The Referee is correct that the pixel size is different between the corrected and uncorrected probes. This is because of the different magnification factor introduced by the corrective microlens, as described in Figure 3E, F. The rationale for showing images in Figure 3AD in pixels rather than microns is the following:

(1) Optical simulations in Figure 1 suggest that a corrective optical element is effective in compensating for some of the optical aberrations in GRIN microendoscopes.

(2) After fabricating the corrective optical element (Figure 2), in Figure 3A-D we conduct a preliminary analysis of the effect of the corrective optical element on the optical properties of the GRIN lens. We observed that the microfabricated optical element corrected for some aberrations (e.g., astigmatism), but also that the microfabricated optical element was

characterized by significant field curvature. This can be appreciated showing distances in pixels.

(3) The observed field curvature and the aspherical profile of the corrected lens prompted us to characterize the magnification factor of the corrected endoscopes as a function of the radial distance. We found that the magnification factor changed as a function of the radial distance (Figure 3E-F) and that pixel size was different between uncorrected and corrected endoscopes. We also observed that, in corrected endoscopes, pixel size was a function of the radial distance (Figure 3E-F).

(4) Once all of the above was established and quantified, we assigned precise pixel size to images of uncorrected and corrected endoscopes and we show all following images of the study (Figure 3G on) using a micron (rather than pixel) scale.

*(3) There seems to be a discrepancy between the performance of the long lenses (8.8 mm) in the different experiments, which should be discussed in the article. For example, the results in Figure 4 show a considerable enlargement of the FOV, whereas the results in Figure 6 show a very moderate enlargement of the distance at which the person's correlation with the first ground truth emitter starts to drop.*

Thanks for raising this point and helping us clarifying data presentation. Images in Figure 4B are average z-projections of z-stacks acquired through a mouse fixed brain slice and they were taken with the purpose of showing all the neurons that could be visualized from the same sample using an uncorrected and a corrected microendoscope. In Figure 4B, all illuminated neurons are visible regardless of whether they were imaged with high axial resolution (e.g.,  $< 10 \mu\text{m}$  as defined in Figure 3J) or poor axial resolution. In contrast, in Figure 6J we evaluated the correlation between the calcium trace extracted from a given ROI and the real activity trace of the first simulated ground truth emitter for that specific ROI. The moderate increase in the correlation for the corrected microendoscope compared to the uncorrected microendoscope (Figure 6J) is consistent with the moderate improvement in the axial resolution of the corrected probe compared to the uncorrected probe at intermediate radial distances (60-100  $\mu\text{m}$  from the optical axis, see Figure 3J). We added a paragraph in the Results section (page 14, lines 8-18) to summarize the points described above.

*a) There is also a significant discrepancy between measured and simulated optical performance, which is not discussed. Optical simulations (Figure 1) show that the useful FOV (defined as the radius for which the size of the PSF along the optical axis remains below  $10 \mu\text{m}$ ) should be at least  $90 \mu\text{m}$  for the corrected microendoscopes of both lengths. However, for the long microendoscopes, Figure 3J shows that the axial resolution at  $90 \mu\text{m}$  is  $17 \mu\text{m}$ . It would be interesting to discuss the origin of this discrepancy: does it depend on the microendoscope used?*

As the Reviewer correctly pointed out, the size of simulated PSFs at a given radial distance (e.g.,  $90 \mu\text{m}$ ) tends to be generally smaller than that of the experimentally measured PSFs. This might be due to multiple reasons:

(1) simulated PSFs are excitation PSFs, i.e. they describe the intensity spatial distribution of focused excitation light. On the contrary, measured PSFs result from the excitation and emission process, thus they are also affected by aberrations of light emitted by fluorescent beads and collected by the microscope.

(2) in the optical simulations, the Zemax file of the GRIN lenses contained first-order aberrations. High-order aberrations were therefore not included in simulated PSFs.

(3) intrinsic variability of experimental measurements (e.g., intrinsic variability of the fabrication process, alignment of the microendoscope to the optical axis of the microscope,

the distance between the GRIN back end and the objective...) are not considered in the simulations.

We added a paragraph in the Discussion section (page 17, lines 9-18) summarizing the abovementioned points.

Are there inaccuracies in the construction of the aspheric corrective lens or in the assembly with the GRIN lens? If there is variability between different lenses, how are the lenses selected for imaging experiments?

The fabrication yield, i.e. the yield of generating the corrective lenses, using molding was ~ 90% (N > 30 molded lenses). The main limitation of this procedure was the formation of air bubbles between the mold negative and the glass coverslip. Molded lenses were visually inspected with the stereoscope and, in case of air bubble formation, they were discarded.

The assembly yield, i.e. the yield of correct positioning of the GRIN lens with respect to the coverslip, was 100 % (N = 27 endoscopes).

We added this information in the Methods at page 29 (lines 1-12), as follows:

“After UV curing, the microlens was visually inspected at the stereomicroscope. In case of formation of air bubbles, the microlens was discarded (yield of the molding procedure: ~ 90 %, N > 30 molded lenses). The coverslip with the attached corrective lens was sealed to a customized metal or plastic support ring of appropriate diameter (Fig. 2C). The support ring, the coverslip and the aspherical lens formed the upper part of the corrected microendoscope, to be subsequently coupled to the proper GRIN rod (Table 2) using a custom-built opto-mechanical stage and NOA63 (Fig. 2C) 7. The GRIN rod was positioned perpendicularly to the glass coverslip, on the other side of the coverslip compared to the corrective lens, and aligned to the aspherical lens perimeter (Fig. 2C) under the guidance of a wide field microscope equipped with a camera. The yield of the assembly procedure for the probes used in this work was 100 % (N = 27 endoscopes). For further details on the assembly of corrected microendoscope see(7)”.

**Reviewer #1 (Recommendations for the authors):**

(1) Page 4, what is meant by 'ad-hoc' in describing software control?

With “ad-hoc” we meant “specifically designed”. We revised the text to make this clear.

(2) It was hard to tell how the PSF was modeled for the simulations (especially on page 34, describing the two spherical shells of the astigmatic PSF and ellipsoids modeled along them). Images or especially videos that show the modeling would make this easier to follow.

Simulated calcium t-series were generated following previous work by our group (Antonini et al., eLife 2020), as stated in the Methods on page 37 (line 5). In Figure 4A of Antonini et al. eLife 2020, we provided a schematic to visually describe the procedure of simulated data generation. In the present paper, we decided not to include a similar drawing and cite the eLife 2020 article to avoid redundancy.

(3) Some math symbols are missing from the methods in my version of the text (page 36/37).

We apologize for the inconvenience. This issue arose in the PDF conversion of our Word document and we did not spot it at the time of submission. We will now make sure the PDF version of our manuscript correctly reports symbols and equations.



*(4) The Z extent of stacks (i.e. number of steps) used to generate images in Figure 4 is missing.*

We thank the Reviewer for the comment and we now revised the caption of Figure 4 and the Methods section as follows:

“Figure 4. Aberration correction in long GRIN lens-based microendoscopes enables highresolution imaging of biological structures over enlarged FOVs. A) jRCaMP7f-stained neurons in a fixed mouse brain slice were imaged using 2PLSM ( $\lambda_{exc} = 920$  nm) through an uncorrected (left) and a corrected (right) microendoscope based on the 6.4 mm-long GRIN rod. Images are maximum fluorescence intensity ( $I$ ) projections of a z-stack acquired with a 5  $\mu$ m step size. Number of steps: 32 and 29 for uncorrected and corrected microendoscope, respectively. Scale bars: 50  $\mu$ m. Left: the scale applies to the entire FOV. Right, the scale bar refers only to the center of the FOV; off-axis scale bar at any radial distance ( $x$  and  $y$  axes) is locally determined multiplying the length of the drawn scale bar on-axis by the corresponding normalized magnification factor shown in the horizontal color-coded bar placed below the image (see also Fig. 3, Supplementary Table 3, and Materials and Methods for more details). B) Same results for the microendoscope based on the 8.8 mm-long GRIN rod. Number of steps: 23 and 31 for uncorrected and corrected microendoscope, respectively”.

We also modified the text in the Methods (page 35, lines 1-2):

“(1024 pixels x 1024 pixels resolution; nominal pixel size: 0.45  $\mu$ m/pixel; axial step: 5  $\mu$ m; number of axial steps: 23-32; frame averaging = 8)”.

*(5) Overall, the text is wordy and a bit repetitive and could be cut down significantly in length without loss of clarity. This is true throughout, but especially when comparing the introduction and discussion.*

We edited the text (Discussion and Introduction), as suggested by the Reviewer.

*(6) Although I don't think it's necessary, I would advise including comparison data with an uncorrected endoscope in the same in vivo preparation.*

We thank the Referee for the suggestion. Below, we list the reasons why we decided not to perform the comparison between the uncorrected and corrected endoscopes in the in vivo preparation:

(1) We believe that the comparison between uncorrected and corrected endoscopes is better performed in fixed tissue (Figure 4) or in simulated calcium data (Figure 5-6), rather than in vivo recordings (Figure 7). In fact, in the brain of living mice motion artifacts, changes in fluorophore expression level, variation in the optical properties of the brain (e.g., the presence of a blood vessel over the FOV) may make the comparison of images acquired with uncorrected and corrected microendoscopes difficult, requiring a large number of animals to cancel out the contributions of all these factors. Comparing optical properties in fixed tissue is, in contrast, devoid of these confounding factors.

(2) A major advantage of quantifying how the optical properties of uncorrected and corrected endoscope impact on the ability to extract information about neuronal activity in simulated calcium data is that, under simulated conditions, we can count on a known ground truth as reference (e.g., how many neurons are in the FOV, where they are, and which is their electrical activity). This is clearly not possible under in vivo conditions.

(3) The proposed experiment requires to perform imaging in the awake mouse with a corrected microendoscope, then anesthetize the animal to carefully remove the corrective microlens using forceps, and finally repeat the optical recordings in awake mice with the uncorrected microendoscope. Although this is feasible (we performed the proposed experiment in Antonini *et al.* eLife 2020 using a 4.1 mm-long microendoscope), the yield of success of these experiments is low. The low yield is due to the fact that the mechanical force applied on top of the microendoscope to remove the corrective microlens may induce movement of the GRIN lens inside the brain, both in vertical and horizontal directions. This can randomly result in change of the focal plane, death or damage of the cells, tissue inflammation, and bleeding. From our own experience, the number of animals used for this experiment is expected to be high.

**Reviewer #2 (Recommendations for the authors):**

*Below, I provide a few minor corrections and suggestions for the authors to consider before final submission.*

*(1) Page 5: when referring to Table 1 maybe add "Table 1 and Methods".*

Following the Reviewer's comment, we revised the text at page 6 (lines 4-5 from bottom) as follows:

“(see Supplementary Table 1 and Materials and Methods for details on simulation parameters)”.

*(2) Page 8: "We set a threshold of 10  $\mu\text{m}$  on the axial resolution to define the radius of the effective FOV (corresponding to the black triangles in Fig. 3I, J) in uncorrected and corrected microendoscopes. We observed an enlargement of the effective FOV area of 4.7 times and 2.3 times for the 6.4 mm-long micro endoscope and the 8.8 mm-long micro endoscope, respectively (Table 1). These findings were in agreement with the results of the ray-trace simulations (Figure 1) and the measurement of the subresolved fluorescence layers (Figure 3AD)." I could not find the information given in this paragraph, specifically:*

*a) Upon examining the black triangles in Figure 3I and J, the enlargement of the effective FOV does not appear to be 4.7 and 2.3 times.*

In Figure 3I, J, black triangles mark the intersections between the curves fitting the data and the threshold of 10  $\mu\text{m}$  on the axial resolution. The values on the x-axis corresponding to the intersections (Table 1, “Effective FOV radius”) represent the estimated radius of the effective FOV of the probes, i.e. the radius within which the microendoscope has spatial resolution below the threshold of 10  $\mu\text{m}$ . The ratios of the effective FOV radii are 2.17 and 1.53 for the 6.4 mm- and the 8.8 mm-long microendoscope, respectively, which correspond to 4.7 and 2.3 times larger FOV (Table 1). To make this point clearer, we modified the indicated sentence as follows (page 10, lines 3-11 from bottom):

“We set a threshold of 10  $\mu\text{m}$  on the axial resolution to define the radius of the effective FOV (corresponding to the black triangles in Fig. 3I, J) in uncorrected and corrected microendoscopes. We observed a relative increase of the effective FOV radius of 2.17 and 1.53 for the 6.4 mm- and the 8.8 mm-long microendoscope, respectively (Table 1). This corresponded to an enlargement of the effective FOV area of 4.7 times and 2.3 times for the 6.4 mm-long microendoscope and the 8.8

mm-long microendoscope, respectively (Table 1). These findings were in agreement with the results of the ray-trace simulations (Figure 1) and the measurement of the subresolved

fluorescence layers (Figure 3A-D)."

*b) I do not understand how the enlargements in Figure 3I and J align with the ray trace simulations in Figure 1, indicating an enlargement of 5.4 and 5.6.*

In Figure 1C, E of the first submission we showed the Strehl ratio of focal spots focalized after the microendoscope, in the object plane, as a function of radial distance from the optical axis of focal spots focalized in the focal plane at the back end of the GRIN rod ("Objective focal plane" in Figure 1A, B), before the light has traveled along the GRIN lens. After reading the Referee's comment, we realized this choice does not facilitate the comparison between Figure 1 and Figure 3I, J. We therefore decided to modify Figure 1C, E by showing the Strehl ratio of focal spots focalized after the microendoscope as a function of their radial distance from the optical axis in the objet plane (where the Strehl ratio is computed), after the light has traveled through the GRIN lens (radial distances are still computed on a plane, not along the curved focal surface represented by the "imaging plane" in Figure 1 A, B). Computing radial distances in the object space, we found that the relative increase in the radius of the FOV due to the correction of aberrations was 3.50 and 3.35 for the 6.4 mm- and the 8.8 mm-long microendoscope, respectively. We also revised the manuscript text accordingly (page 7, lines 6-8):

"The simulated increase in the radius of the diffraction-limited FOV was 3.50 times and 3.35 times for the 6.4 mm-long and 8.8 mm-long probe, respectively (Fig. 1C, E)". We believe this change should facilitate the comparison of the data presented in Figure 1 and Figure 3.

Moreover, in comparing results in Figure 1 and Figure 3, it is important to keep in mind that:

(1) the definitions of the effective FOV radius were different in simulations (Figure 1) and real measurements (Figure 3). In simulations, we considered a theoretical criterion (Maréchal criterion) and set the lower threshold for a diffraction-limited FOV to a Strehl ratio value of 0.8. In real measures, the effective FOV radius obtained from fluorescent bead measurements was defined based on the empirical criterion of setting the upper threshold for the axial resolution to 10  $\mu\text{m}$ .

(2) the Zemax file of the GRIN lenses contained low-order aberrations and not high-order aberrations.

(3) the small variability in some of the experimental parameters (e.g., the distance between the GRIN back end and the focusing objective) were not reflected in the simulations.

Given the reasons listed above, it is expected that the prediction of the simulations do not perfectly match the experimental measurements and tend to predict larger improvements of aberration correction than the experimentally measured ones.

*c) Finally, how can the enlargement in Figure 3I be compared to the measurements of the sub-resolved fluorescence layers in Figures 3A-D? Could the authors please clarify these points?*

When comparing measurements of subresolved fluorescent films and beads it is important to keep in mind that the two measures have different purposes and spatial resolution. We used subresolved fluorescent films to visualize the shape and extent of the focal surface of microendoscopes in a continuous way along the radial dimension (in contrast to bead measurements that are quantized in space). This approach comes at the cost of spatial resolution, as we are using fluorescent layers, which are subresolved in the axial but not in the radial dimension. Therefore, fluorescent film profiles are not used in our study to extract relevant quantitative information about effective FOV enlargement or spatial resolution of corrected microendoscopes. In contrast, to quantitatively characterize axial and lateral

resolutions we used measurements of 100 nm-diameter fluorescent beads (therefore subresolved in the x, y, and z dimensions) located at different radial distances from the center of the FOV, using a much smaller nominal pixel size compared to the fluorescent films (beads, lateral resolution: 0.049  $\mu\text{m}/\text{pixel}$ , axial resolution: 0.5  $\mu\text{m}/\text{pixel}$ ; films, lateral resolution: 1.73  $\mu\text{m}/\text{pixel}$ , axial resolution: 2  $\mu\text{m}/\text{pixel}$ ).

*(3) On page 15, the statement "significantly enlarge the FOV" should be more specific by providing the actual values for the increase. It would also be good to mention that this is not a xy lateral increase; rather, as one moves further from the center, more of the imaged cells belong to axially different planes.*

The values of the experimentally determined FOV enlargements (4.7 times and 2.3 times for 6.4 mm- and 8.8 mm-long microendoscope, respectively) are provided in Table 1 and are now referenced on page 10. Following the Referee's request, we added the following sentence in the discussion (page 18, lines 10-14) to underline that the extended FOV samples on different axial positions because of the field curvature effect:

"It must be considered, however, that the extended FOV achieved by our aberration correction method was characterized by a curved focal plane. Therefore, cells located in different radial positions within the image were located at different axial positions and cells at the border of the FOV were closer to the front end of the microendoscope".

*(4) On page 36, most of the formulas appear to be corrupted. This may have occurred during the conversion to the merged PDF. Please verify this and check for similar problems in other equations throughout the text as well.*

We apologize for the inconvenience. This issue arose in the PDF conversion of our Word document and we did not spot it upon submission. We will now make sure the PDF version of our manuscript correctly reports symbols and equations.

*(5) In the discussion, the authors could potentially add comments on how the verified performance of the corrective lenses depends on the wavelength and mention the range within which the wavelength can be changed without the need to redesign a new corrective lens.*

Following this comments and those of other Reviewers, we explored the effect of changing wavelength on the Strehl ratio using new Zemax simulations. We found that the Strehl ratio remains  $> 0.8$  within  $\pm$  at least 10 nm from  $\lambda = 920$  nm (new Supplementary Figure 1A-D, left panels), which covers the limited bandwidth of our femtosecond laser. Moreover, these simulations demonstrate that, on a much wider wavelength range (800 - 1040 nm), high Strehl ratio is obtained but at different z planes (new Supplementary Figure 1A-D, right panels). These new results are now described on page 7 (lines 8-10).

*(6) Also, they could discuss if and how the corrective lens could be integrated into fiberscopes for freely moving experiments.*

Following the Referee's suggestion, we added a short text in the Discussion (page 21, lines 4-7 from bottom). It reads:

"Another advantage of long corrected microendoscopes described here over adaptive optics approaches is the possibility to couple corrected microendoscopes with portable 2P microscopes(42-44), allowing high resolution functional imaging of deep brain circuits on an enlarged FOV during naturalistic behavior in freely moving mice".

*(7) Finally, since the main advantage of this approach is its simplicity, the authors should also comment on or outline the steps to follow for potential users who are interested in using the corrective lenses in their systems.*

Thanks for this comment. The Materials and Methods section of this study and that of Antonini et al. eLife 2020 describe in details the experimental steps necessary to reproduce corrective lenses and apply them to their experimental configuration.

**Reviewer #3 (Recommendations for the authors):**

*(1) Suggestions for improved or additional experiments, data, or analyses, and Recommendations for improving the writing and presentation:*

*See Public Review.*

Please see our point-by-point response above.

*(2) Minor corrections on text and figures: a) Figure 6A: is the fraction of cells expressed in %?*

Author response: yes, that is correct. Thank you for spotting it. We added the “%” symbol to the y label.

*b) Figure 8A, left: The second line is blue and not red dashed. In addition, it could be interesting to also show a line corresponding to the 0 value.*

Thank you for the suggestions. We modified Figure 8 according to the Referee’s comments.

*c) Some parts of equation (1) and some variables in the Material and Methods section are missing*

We apologize for the inconvenience. This issue arose in the PDF conversion of our Word document and we did not spot it upon submission. We will now make sure the PDF version of our manuscript correctly reports symbols and equations.

*d) In the methods, the authors mention a calibration ruler with ticks spaced every 10  $\mu\text{m}$  along two orthogonal directions and refer to the following product: 4-dot calibration slide, Cat. No. 1101002300142, Motic, Hong Kong. However, this product does not seem to correspond to a calibration ruler.*

We double check. The catalog number 1101002300142 is correct and product details can be found at the following link:

[https://moticmicroscopes.com/products/calibration-slide-4-dots-1101002300142?srsId=AfmBOorGYx9PcXtAlIMmSs\\_tEpxS4nX21qIcV8Kfn4qGwizQK3LYOQn3](https://moticmicroscopes.com/products/calibration-slide-4-dots-1101002300142?srsId=AfmBOorGYx9PcXtAlIMmSs_tEpxS4nX21qIcV8Kfn4qGwizQK3LYOQn3)

<https://doi.org/10.7554/eLife.101420.2.sa0>